Momentum Funds

AQR Case Study & Empirical Analysis of the Momentum Premium (1927–2023)

Asset Allocation & Investment Strategies: ELE Group 3

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Abstract

Based on AQR's Momentum Funds (HBS Case 9-209-119), the GMO White Paper "Momentum — A Contrarian Case for Following the Herd" (Hancock, 2010), and Fama-French factor data.

This notebook begins by doing a deep dive into AQR to decide whether they should deploy a momentum based strategy and the specific design requirements and limitations of such a strategy. We then systematically examines historical evidence on the Fama-French MOM (momentum) factor using **both** the factor data and the 49 industry portfolio data to form an informed opinion on whether momentum returns over the next decade will be significantly greater than zero, significantly less than zero, or approximately zero. From this we examine the correlation betwen momentum and value and decide if intrducung a new momentum index based off industry specific momentum has merit.

Section 1 - AQR Case Study

Question 1

Should AQR Launch the Momentum Funds?

Yes, AQR should launch the Momentum Funds. The decision is supported by the empirical durability of the momentum premium, the strategic opportunity presented by a retail market where the strategy does not yet exist as a labelled product, and the diversification properties of momentum in multi-factor portfolios. The principal risks, namely crash dynamics and implementation costs, are acknowledged below but do not change this conclusion.

The Empirical Case

The Fama-French UMD factor, a long-short portfolio of the top 30th percentile of past-return stocks against the bottom 30th percentile (skipping the most recent month), provides the standard benchmark. Table 1 reports statistics computed from monthly data over January 1927 through December 2008 (n = 984 observations), the information set available to AQR at the time of the decision.

Table 1: Summary Statistics - Fama-French MOM Factor (1927-2008) [1] (Authors' Calculations)

Metric	Value	Metric	Value
Annualised mean return	9.27%	Three-factor alpha (ann.)	13.35%
Annualised σ	16.29%	Alpha t-statistic	≈ 8.5
Sharpe ratio	0.569	3-factor market beta	-0.21
t-statistic (mean = 0)	5.15	3-factor HML loading	-0.48
% of years positive	79.3%	Monthly skewness	-3.08
Monthly excess kurtosis	29.31		

The mean return of 9.27% per annum is economically large and highly significant (t = 5.15), positive in 79.3% of calendar years. An OLS three-factor regression yields an annualised alpha of 13.35% (t $\approx$ 8.5), indicating the premium is not explained by market, size, or value exposures. The three-factor market beta is modestly negative (-0.21) while the market correlation is near zero (-0.03), so momentum has little systematic exposure to broad market risk; the strongly negative HML loading (-0.48) indicates meaningful anti-value tilt, consistent with the -0.30 correlation with HML reported in Exhibit 5.

A key question for AQR is whether the premium has eroded since Jegadeesh and Titman's 1993 publication. The case data suggest not. Using the case's own UMD series (constructed with 30th/70th percentile breakpoints, as reported in the case text rather than the decile-based series in Table 1), the post-publication annual average over 1993-2008 was 11.48% per annum, above the pre-publication average of 10.79%. Table 2 disaggregates performance by decade.

Table 2: Decade-by-Decade MOM Performance (1927-2008) [1] Authors' calculations. 1920s omitted (partial decade in sample).

Decade	Ann. Mean	Ann. σ	Sharpe	t-stat
1930s	0.16%	32.85%	0.01	0.02
1940s	6.59%	9.41%	0.70	2.22
1950s	10.74%	7.83%	1.37	4.34
1960s	11.09%	9.37%	1.18	3.74
1970s	9.88%	13.07%	0.76	2.39
1980s	9.19%	12.38%	0.74	2.35
1990s	13.61%	11.15%	1.22	3.86
2000-2008	8.37%	20.73%	0.40	1.21

Note: 1930s Sharpe and t-statistic are depressed by extreme 1932 reversal events.

The premium is positive in seven of the eight post-1920s decades. The exception is the 1930s (0.16%, t = 0.02), driven by two months in 1932 in which prior losers strongly outperformed winners, collapsing the Winners-minus-Losers return at the Depression trough (August 1932: -52.05%; July 1932: -45.02%). The 2000-2008 period shows an annualised return of 8.37% (t = 1.21): below the long-run average and statistically insignificant, partly reflecting the tech-bubble aftermath and elevated volatility over the period, though still positive. Importantly, this 2000-2008 figure represents the actual information set available to AQR at the time of the decision; the spring 2009 reversal had not yet occurred.

Table 3 reinforces the monotonic relationship between past-return rank and future return that underlies the entire strategy, using all ten decile portfolios from the Fama-French data (Exhibit 3/4). The pattern is striking: average annual returns rise nearly monotonically from 2.42% in the loser decile to 20.42% in the winner decile, a spread of nearly 18 percentage points, and the long-short portfolio earns this spread with a near-zero market beta of -0.08.

Table 3: Average Annual Returns Across All Ten Momentum Deciles (1927-2008) [1]

Decile	Label	Ann. Return	Ann. σ	CAPM Beta	CAPM Alpha
1	Losers	2.42%	30.35%	1.28	-7.34%
2		7.51%	–	–	–
3		7.95%	–	–	–
4		10.22%	–	–	–
5		10.05%	–	–	–
6		11.17%	–	–	–
7		12.47%	–	–	–
8		14.75%	–	–	–
9		15.60%	–	–	–
10	Winners	20.42%	28.52%	1.20	11.23%
10 - 1	Long-Short	17.99%	21.21%	-0.08	18.57%

What the Long-Only Constraint Costs

The mutual fund format restricts AQR to the long leg of the strategy. From Table 3 above, the winner decile (decile 10) delivered an average annual value-weighted return of 20.42% with a standard deviation of 28.52% over 1927-2008. More directly relevant to the proposed funds, the case reports that the AQR large-cap momentum index outperformed the Russell 1000 by 210 basis points per annum since 1980 and the Russell 1000 Growth index by 320 basis points; these figures incorporate the index's quarterly rebalancing frequency and large-cap liquidity screen, making them the most appropriate benchmark for expected fund performance.

On implementation costs: momentum strategies require high portfolio turnover. Assuming turnover of approximately 300-500% per year, round-trip transaction costs of ~0.5% per turnover, and a management fee of ~0.65%, the total drag is approximately 2-3% per annum. The gross winner premium is sufficient to sustain a positive net excess return under this scenario. Forgoing the short leg reduces the strategy's overall return potential, but the long-only fund remains a potentially superior expected-return alternative to a passive allocation, conditional on managing implementation costs and accepting the higher volatility (Exhibit 5: market standard deviation 21.01% versus 28.52% for the winner decile) and crash risk discussed below.

Strategic Positioning and Diversification Value

Table 4 places the momentum factor in the context of the other Fama-French factors, using the full-sample summary statistics from Exhibit 5 of the case. The comparison highlights two properties that are central to AQR's product rationale: momentum has the highest mean return of any single factor, and it has the lowest (most negative) correlation with both the market and value, giving it genuine diversification value that no other factor in the table provides.

Table 4: Fama-French Factor Summary Statistics and Cross-Correlations (1927-2008) [1]

Factor	Ann. Mean	Ann.	Sharpe	Corr w/ Mkt-RF	Corr w/ HML	Corr w/ UMD
Mkt-RF	7.64%	21.01%	0.36	1.00	0.11	-0.03
SMB	3.56%	14.37%	0.25	0.41	0.08	-0.09
HML (Value)	5.14%	14.04%	0.37	0.11	1.00	-0.30
UMD (Momentum)	10.92%	12.34%	0.88	-0.03	-0.30	1.00

Source: Fama-French Data Library, Exhibit 5. Sharpe ratios calculated using reported means and standard deviations; risk-free rate effects already removed (factors reported as excess returns).

Momentum's Sharpe ratio of 0.88 is more than double that of either value (0.37) or the market (0.36) over the full sample, and its negative correlations with both the market (-0.03) and HML (-0.30) mean that it adds diversification value unavailable from any other factor in the table.

Momentum does not exist as a mainstream, labelled retail mutual fund product in early 2009, which gives AQR a clear first-mover opportunity to establish a track record and build advisor distribution relationships ahead of potential competitors. AQR's experience running momentum strategies within its hedge funds also represents a meaningful implementation advantage: executing a high-turnover factor strategy cost-efficiently in a retail vehicle requires operational infrastructure and market-impact expertise that takes time to develop, and which confers a structural edge over later entrants.

Risks

The strategy carries two principal risks. The monthly return distribution exhibits skewness of -3.08 and excess kurtosis of 29.31, indicating a fat left tail with exposure to severe episodic drawdowns. The 1932 data illustrates the magnitude: losses exceeding 50% in a single month occurred when prior losers recovered sharply and the Winners-minus-Losers construction collapsed. In an open-end vehicle, such episodes create the additional risk of forced redemptions at the point of maximum loss. The long-only format does provide partial mitigation, as the most extreme crash dynamics in a long-short momentum strategy are driven by the short-side loser recovery; this mechanism is absent in the mutual fund. However, sharp market reversals can still produce significant losses through winner underperformance, so crash risk is reduced rather than eliminated. The second risk is premium attrition through transaction costs: momentum's high turnover makes execution efficiency the primary determinant of net performance, and poor implementation discipline can erode the gross premium substantially. Both risks are material, but both are manageable and neither constitutes a structural argument against launching the product.

Conclusion

AQR should launch the Momentum Funds. Using data through December 2008, the MOM factor shows a mean return of 9.27% per annum ($t = 5.15$) and a three-factor alpha of 13.35% ($t \approx 8.5$), among the most robust premia in the empirical asset-pricing literature. The premium is positive in 79.3% of calendar years and held above its pre-1993 level in the post-publication period. The annual correlations of -0.03 with the market and -0.30 with HML (Exhibit 5) give the strategy genuine diversification value in multi-factor portfolios. AQR's first-mover position and implementation experience make it well placed to preserve the net premium in a retail format. The launch should proceed, with conservative fee-setting and explicit communication to investors regarding the strategy's crash characteristics.

Question 2

How important do you think it is that AQR establish a momentum index at the same time they launched their funds?

Establishing a momentum index at launch was strategically critical for AQR. The index transformed momentum from an academic anomaly into a benchmarked, investable style compatible with U.S. mutual fund regulation and retail distribution. It provided the analytical, commercial, and legal infrastructure without which the momentum funds would have lacked regulatory clarity, strategic coherence, and benchmarking discipline.

Regulatory and Disclosure Rationale

U.S. mutual fund advertising rules do not permit managers to market products using purely hypothetical or back-tested performance. Momentum's principal selling point is its robust long historical track record, documented in academic studies and factor libraries. Without some alternative mechanism, AQR would have been unable to show retail investors the very evidence that justifies the strategy.

By establishing a formal momentum index at launch and having it calculated by an independent agent (Standard & Poor's), AQR created a compliant channel through which a long pre-launch performance history of a rules-based momentum strategy could be published. The index can display volatility, average returns, and drawdowns over many decades, yet it is not presented as the fund's track record, so it fits within mutual fund marketing constraints. The index bridges the gap between academic back-tests and permitted disclosure, enabling AQR to communicate the empirical case for momentum to advisors and retail clients.

From Academic Factor to Investable Benchmark

The standard academic benchmark for momentum is the Fama-French UMD factor: a long-short portfolio of high-past-return stocks against low-past-return stocks, rebalanced monthly across the whole stock universe. This is a research construct rather than a practical benchmark for a retail mutual fund. A retail momentum fund faces very different constraints – it must be diversified, liquid, long-only, and sensitive to transaction costs, with no shorting or leverage. If AQR simply pointed to UMD, investors would be comparing a constrained long-only mutual fund to a hypothetical, unconstrained long-short factor, producing misleading performance expectations and criticism whenever the fund diverged from UMD.

Designing a proprietary momentum index at launch allowed AQR to align the benchmark with those real-world constraints. The AQR indices are long-only, built on defined liquid universes, select the top-momentum segment based on past 12-to-2-month returns, are value-weighted, and rebalance at quarterly frequency. The benchmark thereby respects the same liquidity, regulatory frictions, and turnover the fund itself must respect. That alignment makes comparisons between fund and index economically meaningful. Table 5 summarises the key contrasts.

Table 5: Academic UMD Factor vs. AQR Long-Only Momentum Index

Dimension	Fama-French UMD	AQR Momentum Index
Direction	Long-short	Long-only
Rebalancing frequency	Monthly	Quarterly
Universe	All listed stocks	Largest 1,000 liquid stocks (large-cap)
Weighting	Equal or value	Value-weighted (cap-weighted)
Shorting / leverage	Yes	No
Purpose	Academic research factor	Replicable inside a 1940-Act mutual fund
Disclosure use	Academic publication	Third-party calculated; publishable pre-launch

Investor Education and Expectation Management

Momentum buys recent winners that many investors perceive as overpriced and is prone to sharp reversals and occasional crashes. Without a transparent benchmark, there is a high risk that investors misinterpret normal factor drawdowns as evidence of manager incompetence and redeem at precisely the wrong time.

A long-history, rules-based index stabilises expectations in several ways. It allows advisors and clients to see the full historical pattern of a momentum strategy, including both long-run outperformance and episodes of severe underperformance. It provides a way for investors to distinguish between the factor being in a drawdown and the manager underperforming the factor. It also clarifies that high volatility and occasional large drawdowns are intrinsic features of the momentum style, not uniquely caused by AQR's implementation choices. This expectation management role is particularly important in an open-end mutual fund, where capital is not locked up and investors can redeem daily. A credible index reduces the risk of behavioural overreaction and forced outflows during temporary momentum reversals.

Implementation Discipline and Performance Attribution

Once the index exists, it becomes the reference portfolio that structures AQR's implementation decisions. The index encodes a transparent baseline for how to run a long-only momentum strategy under mutual fund constraints. Any deviation by the fund can then be interpreted and evaluated against this baseline.

Key implementation questions, including how often to rebalance, how aggressively to trade stocks near the threshold, how to balance transaction costs against tracking error, and how to manage taxes, can be framed as active choices relative to the index. This makes performance attribution clear: the index captures the return to pure momentum exposure under agreed constraints, while any excess or shortfall of fund performance relative to the index reflects implementation skill, after trading, tax management, and other enhancements. Without such an index, AQR would be judged only relative to broad market or growth benchmarks, which would conflate style returns with manager decisions and make it much harder to explain outcomes to sophisticated investors.

Strategic Positioning and Category Creation

By launching the index contemporaneously with the funds, AQR moved from being a user of the momentum concept to being a definer of the momentum category in the retail market. The index gives AQR several strategic advantages: it allows AQR to specify the practical definition of long-only momentum (universe, ranking window, weightings, rebalance rules) that other market participants may adopt by default; with a branded third-party-calculated index, AQR can position itself as the reference provider of momentum benchmarks, with potential to license the index to other managers; and it strengthens AQR's brand as a research-driven quantitative firm that not only runs strategies but also sets factor standards, analogous to how established index providers shaped value and growth as investment categories.

Conclusion

The momentum index was not an add-on but a necessary foundation for the product. It enabled AQR to communicate momentum's long history within regulatory limits, provided an investable benchmark tailored to mutual fund frictions, disciplined implementation and attribution, managed investor expectations in a crash-prone strategy, and positioned AQR as the architect of the momentum asset class in the retail channel. Without establishing the index at launch, the momentum funds would have lacked the strategic, analytical, and regulatory infrastructure required for a credible and scalable offering.

Question 3

What do you think are the appropriate benchmarks for the AQR Momentum Funds? Do you believe the net performance of the Funds will exceed those benchmarks? Why or why not?

Because the Momentum Funds are designed to capture the momentum factor under long-only mutual fund constraints, a single benchmark is insufficient to evaluate their performance. The appropriate framework separates two distinct questions: does the fund faithfully implement the momentum factor, and does the momentum factor add value relative to conventional alternatives? Answering both questions requires a multi-layered benchmarking approach.

Primary Benchmark: The AQR Long-Only Momentum Index

The AQR large-cap momentum index is the most economically appropriate primary benchmark. It is constructed from the same universe (the largest 1,000 U.S. stocks), uses the same 12-to-2-month return formation window, applies the same value-weighting and quarterly rebalancing, and is explicitly designed for the long-only, open-end mutual fund context in which the fund operates. Comparing the fund to this index isolates implementation skill: any deviation between fund and index reflects decisions about trade timing, boundary stock handling, turnover control, and tax efficiency rather than the performance of the momentum factor itself.

Because the index is engineered to replicate the same frictions and constraints the fund faces, persistent fund outperformance versus the index would signal genuine added value from active implementation choices, while persistent underperformance would indicate that transaction costs, poor trade execution, or other frictions are eroding the gross factor premium. Since both the fund and the index are subject to the same structural constraints, this comparison is the cleanest available test of whether AQR's operational and quantitative capabilities are adding or subtracting value net of costs.

Secondary Benchmarks: Broad Market and Style Indices

Retail investors and financial advisors evaluate fund performance primarily against familiar market indices. The case provides two directly relevant comparisons: the AQR large-cap momentum index outperformed the Russell 1000 by 210 basis points per annum since 1980 and the Russell 1000 Growth index by 320 basis points per annum over the same period. These figures are computed from the index rather than the live fund, but they represent the best available estimate of the gross premium that a well-implemented long-only momentum strategy can be expected to deliver above conventional passive alternatives.

The Russell 1000 Growth is arguably the more meaningful secondary benchmark. Momentum portfolios systematically overweight recent high-returners, which tend to be growth-oriented stocks with high valuations and strong recent price appreciation. The portfolio's characteristics therefore resemble a growth allocation more closely than a broad market allocation, making the Russell 1000 Growth the more natural style comparison. The 320-basis-point average outperformance over the 1980-to-2009 period is large enough to survive substantial implementation costs and still represent a meaningful advantage to investors, provided those costs are contained.

Expected Net Performance Relative to the Benchmarks

The direct answer is: positive gross premium versus both Russell benchmarks, but after costs the net outcome is uncertain versus the Russell 1000 Growth and likely negative versus the Russell 1000, and structurally negative versus the AQR Momentum Index.

Table 6 sets out the net performance logic by decomposing the gross premium into its components and mapping the residual against each benchmark.

Table 6: Net Performance Waterfall – AQR Large-Cap Momentum Fund vs. Benchmarks

Component	vs. Russell 1000	vs. Russell 1000 Growth	vs. AQR MOM Index
Gross premium (index history, ann.)	+210 bps	+320 bps	0 bps (by definition)
Less: Transaction costs (300-500% turnover × ~0.5%)	-150 to -250 bps	-150 to -250 bps	-150 to -250 bps
Less: Management fee (~0.65%)	-65 bps	-65 bps	-65 bps
Less: Tax drag (short-term gains, est.)	-30 to -50 bps	-30 to -50 bps	-30 to -50 bps
Estimated net outperformance	**-155 to -35 bps**	**-45 to +75 bps**	**-365 to -245 bps**
Verdict	**Negative after costs**	**Marginal / uncertain**	**Persistent lag expected**

Source: Authors' calculations. Gross premia from AQR index history (case data). Transaction cost estimates based on assumed turnover and round-trip costs as discussed in Question 1. Tax drag estimated for a taxable investor holding a high-turnover fund through a financial advisor channel.

Table 6 reveals a sobering arithmetic reality: under the cost assumptions used throughout this analysis (300-500% annual turnover at ~0.5% round-trip, a 0.65% management fee, and 30-50 bps of tax drag), net outperformance is negative in all scenarios except the very best-case execution against the Russell 1000 Growth. The 320-basis-point gross premium versus the Russell 1000 Growth is the only column where breakeven is even plausible, and only at the low end of the cost range. Against the Russell 1000 the math is more difficult still, producing a net result of -155 to -35 basis points. This underscores the central message of the case: implementation quality is not a secondary consideration but the primary determinant of whether the fund delivers value to investors. Against the AQR Momentum Index, the fund will structurally lag by the full cost of implementation, since the index represents the same gross exposure without real-world frictions.

Key Risks to Expected Outperformance

Several structural risks could erode the gross premium. High turnover is the most direct threat: even modest increases in market impact costs compound significantly at 300-500% annual turnover. Crowding represents a longer-run concern, as widespread adoption of momentum strategies across institutional investors compresses the spread between winners and losers and reduces the extractable premium. Finally, momentum crashes can produce drawdowns of sufficient magnitude that investors redeem at the trough, crystallising losses and forcing the fund to sell into an adverse market. These risks do not make outperformance unlikely over a sufficiently long horizon, but they do constrain the reliability and timing of that outperformance, which is why benchmark selection and investor communication are closely linked.

Conclusion

AQR should use the proprietary long-only momentum index as the primary benchmark to measure implementation quality and attribution, and the Russell 1000 Growth as the secondary benchmark to communicate the strategy's value-added to retail investors. As Table 6 shows, the fund's gross premium versus the Russell 1000 Growth (320 bps) provides the tightest path to a positive net outcome, achievable only under efficient, low-cost execution. Against the Russell 1000, the math is unlikely to be positive after realistic costs. Outperformance relative to the AQR Momentum Index itself is unlikely to be consistently positive – the index already captures most of the achievable momentum premium under the same constraints the fund faces, and the fund will structurally lag it by the cost of implementation.

Question 4

Does momentum make an attractive product for mutual fund investors? Are other quantitative strategies attractive mutual fund products? If so, which?

Momentum can be an attractive mutual fund product, though its suitability depends critically on how its role within a broader portfolio is defined and communicated. Its empirical strength, distinct factor exposure, and diversification properties make it a compelling component of a style-allocation framework, even if structural features of the mutual fund format constrain the degree to which the theoretical premium can be captured.

The Empirical Appeal of Momentum

A substantial body of empirical evidence, summarised in Question 1, demonstrates that momentum has generated persistent excess returns across long time horizons and multiple markets. Even after controlling for market, size, and value risk, momentum portfolios produce statistically significant abnormal returns, and the premium shows no meaningful erosion in the post-1993 post-publication period. This persistence is a prerequisite for an investable strategy: a premium that disappears once widely known provides no commercial rationale for a product launch.

Momentum also represents a genuinely distinct style exposure rather than a repackaging of existing factors. As Table 4 in Question 1 shows, UMD has a correlation of -0.03 with the market factor and -0.30 with HML over 1927-2008, and a Sharpe ratio of 0.88 that is more than double that of any other individual factor. The negative correlation with value is particularly useful: periods of relative weakness in value strategies have historically corresponded with stronger momentum performance, and vice versa. This complementarity provides a strong rationale for incorporating momentum within multi-factor fund offerings.

Accessibility and Conceptual Clarity

From a product design perspective, momentum rests on relatively intuitive economic reasoning. The practice of allocating capital to recent winners while avoiding recent losers can be articulated clearly to financial advisors and retail investors. In contrast to more complex quantitative approaches that rely on opaque signals or derivative-based structures, momentum is comparatively transparent and accessible. The existence of a clean, rules-based index further aids communication by giving advisors a simple reference point.

Empirical evidence also indicates that momentum effects are observed across asset classes, including bonds, currencies, and commodities. This cross-asset robustness supports the view that momentum reflects a systematic return pattern rather than a narrow statistical anomaly confined to equities alone, which strengthens the commercial narrative.

Cyclicality and Crash Risk

Momentum strategies exhibit marked cyclicality over time. As documented in Question 1, the strategy's monthly return distribution has skewness of -3.08 and excess kurtosis of 29.31, generating episodic crashes that can exceed 50% in a single month. The mechanism is partly attributable to shifts in the beta exposure of prior losers: following market downturns, stocks categorised as past losers often exhibit elevated market beta, so a momentum strategy that underweights these securities incurs losses during sharp recoveries. This asymmetric beta exposure generates pronounced short-term drawdowns that are fundamentally different from the steady underperformance associated with value risk.

Such dynamics indicate that momentum should not be presented to retail investors as a low-risk diversifier. Its episodic losses are large, rapid, and concentrated, and occur precisely when investor sentiment is most fragile. The open-end structure of a mutual fund exacerbates this risk, since daily liquidity requirements mean that investor redemptions can force asset sales into an already declining market.

Other Quantitative Strategies as Mutual Fund Products

Several quantitative factor strategies translate more naturally into the mutual fund structure and are directly relevant to AQR's broader product strategy. Table 7 compares the key dimensions of the main candidates using data from Exhibits 3 and 5 and the wider Fama-French factor literature.

Table 7: Quantitative Factor Strategies as Mutual Fund Products – Comparative Assessment based off period 1927-2008

Strategy	Factor	Ann. Premium	Typical Turnover
Momentum	UMD	10.92%	Very high
Value	HML	5.14%	Low
Market (passive)	Mkt-RF	7.64%	Very low
Size	SMB	3.56%	Low-moderate
Quality / Profitability		~3-5% est.	Low
Low Volatility		~3-5% est.	Low-moderate

Strategy	Long-Only Premium Retention	Crash Risk	MF Suitability
Momentum	Moderate	Very high	Complementary / diversifier
Value	High	Low-moderate (positive skew)	Core holding
Size	High	Low	Niche / small-cap tilt
Quality / Profitability	High	Low	Core / defensive
Low Volatility	High	Low	Defensive / income

Source: Ann. premia for UMD, HML, Mkt-RF, SMB from Fama-French Data Library (Exhibit 5). Quality and low-volatility premia estimated from academic literature (Novy-Marx 2013; Frazzini and Pedersen 2014); not separately reported in the case data.

The table reveals a clear hierarchy for the mutual fund context. Value and quality are the strongest candidates for core holdings: their premia survive the long-only constraint with minimal loss, their turnover is low enough to keep implementation costs manageable, and their drawdown profiles are less severe than momentum's. Momentum occupies a distinct niche as a high-returning, low-correlated diversifier that cannot easily serve as a core holding given its crash characteristics and turnover, but adds genuine portfolio efficiency when combined with value. Size (SMB) offers a modest premium with low crash risk but limited commercial appeal as a standalone product. Low-volatility strategies are attractive for defensive mandates and income-oriented investors.

Conclusion

Momentum can be considered an attractive mutual fund product provided that its role is clearly defined and its risks are appropriately communicated. As Table 7 shows, its combination of the highest single-factor premium (10.92% p.a.), a Sharpe ratio of 0.88, and a strongly negative correlation with value is unmatched by any other factor strategy. However, its turnover, crash risk, and partial premium loss under the long-only constraint mean it is most appropriately positioned as a complementary style allocation alongside value or quality, rather than as a standalone core holding.

Question 5

How important do you think the constraint is that the funds are long-only and do not allow short positions? Is this likely to be more of a concern for momentum than other quantitative strategies?

The long-only constraint is a meaningful limitation for AQR's Momentum Funds, but it is not a fatal one. The constraint reduces expected excess return and increases residual equity market exposure, but the empirical record shows that a well-implemented long-only momentum strategy can still generate positive and economically meaningful alpha relative to passive benchmarks. The more important question is not whether momentum survives the long-only constraint, but how much of the unconstrained premium is lost, and whether that loss is larger for momentum than for other quantitative strategies.

How the Constraint Affects Momentum's Return Potential

In the academic implementation of momentum, the return premium is generated from both sides of the trade: winners outperform and losers underperform. From Table 3 in Question 1, the loser decile (decile 1) produced an average annual return of just 2.42% against the full market's 7.64%, while the winner decile (decile 10) produced 20.42%. A long-short strategy captures the spread between these two groups; a long-only strategy captures only the winner leg. This means that the long-only fund cannot benefit from the loser underperformance, which over the full 1927-2008 period represented approximately half of the gross 10-minus-1 spread of 17.99% per annum.

However, the winner premium in isolation is still large. Decile 10's 20.42% average annual return substantially exceeds the broad market's 7.64%, providing a meaningful expected excess return even without the short book. Moreover, the AQR large-cap momentum index, which is a long-only implementation, outperformed the Russell 1000 by 210 basis points and the Russell 1000 Growth by 320 basis points since 1980, confirming that the long-only format preserves economically meaningful excess return. Table 8 quantifies the key differences between the constrained and unconstrained implementations.

Table 8: Long-Short vs. Long-Only Momentum – Key Metric Comparison

Metric	Long-Short MOM (UMD)	Long-Only MOM (Decile 10)	Difference / Comment
Ann. mean return	17.99% (D10-D1 spread, Exhibit 3)	20.42% (gross winner, Exhibit 3)	D10-D1 spread; UMD factor (30/70 breakpoints) earns 10.92%
CAPM alpha (ann.)	18.57%	11.23%	L/S alpha ~65% higher after market adjustment
Market beta	-0.08	1.20	Long-only fully exposed to equity market
Correlation with Mkt-RF	$ -$0.03	$ +$0.75-0.85	Dramatically higher market exposure long-only
Correlation with HML	-0.30	$ -$0.10 to -0.15	Anti-value tilt reduced long-only
Crash exposure	Full (both legs amplify)	Partial (winner leg only)	Long-only avoids loser-recovery crash mechanism
Regulatory viability	Hedge fund / institutional	Mutual fund compatible	Long-only enables retail distribution

Source: Long-short metrics from Exhibit 3/5. Long-only CAPM alpha and beta from Exhibit 3. Market correlation and HML correlation for long-only estimated from factor structure; exact values not separately reported in case exhibits for decile 10 alone.

The table makes the trade-off concrete: the long-only format eliminates the near-zero market beta of the long-short strategy and reduces the CAPM alpha from 18.57% (D10-D1 long-short, Exhibit 3) to 11.23% per annum for the winner decile – a meaningful but not catastrophic loss. The reward is regulatory viability and retail distribution access that the long-short strategy cannot achieve.

The Constraint Is More Binding for Momentum Than for Other Strategies

The severity of the long-only constraint is not uniform across quantitative factor strategies, and there are good reasons to expect it to be more binding for momentum than for value, quality, or low-volatility. The value premium in the HML factor is partly driven by the outperformance of the long leg (cheap stocks), which survives well in a long-only format. The same applies to quality and low-volatility strategies: the long-only portfolio already represents a coherent and meaningful factor tilt even without the corresponding short, as Table 7 in Question 4 shows.

For momentum, by contrast, the short leg contributes substantially to the strategy's theoretical performance and, perhaps more importantly, to its risk-reduction properties. A long-short momentum portfolio has low market correlation because the long and short positions partially offset in market downturns. Removing the short book eliminates this hedge and also removes the return contribution from loser underperformance. Additionally, the most severe momentum crashes are driven primarily by loser recovery: when prior losers rally sharply in market rebounds, a long-short momentum fund suffers on the short side. While a long-only fund avoids this specific mechanism, it also cannot benefit from holding short positions during the many more normal periods when losers continue to underperform.

Furthermore, the practical benefits of a long-short structure, including the ability to express genuine negative views on low-momentum stocks and to reduce overall portfolio volatility through short-side hedging, cannot be replicated within the mutual fund vehicle. This makes the constraint qualitatively more restrictive for momentum than for factor strategies whose long side already captures most of the available premium.

Implementation Priorities Under the Long-Only Constraint

Given the binding nature of the constraint, AQR's implementation priorities should focus on maximising the efficiency with which the long leg is captured. This means minimising unnecessary turnover by using well-calibrated boundary rules for stocks near the threshold, incorporating short-term reversal and other complementary signals to improve trade timing, and managing tax efficiency to reduce the after-tax impact of the strategy's inherently high turnover. Each of these measures represents a way to recover, at the margins, some of the efficiency that the short book would otherwise provide.

Conclusion

The long-only constraint is a significant but manageable restriction for AQR. As Table 8 shows, it reduces CAPM alpha from approximately 18.57% to 11.23% per annum and raises market beta from near zero to approximately 1.20, meaningfully reducing the strategy's diversification properties relative to the unconstrained version. This constraint is more binding for momentum than for most other quantitative strategies, because the short leg of a momentum portfolio contributes more substantially to both the return premium and the risk management properties of the unconstrained strategy. Nevertheless, the strong historical performance of the winner decile and the AQR index's documented outperformance versus conventional benchmarks confirm that the long-only format does not render the strategy commercially unviable, provided that implementation is efficient.

Question 6

If AQR launches the Momentum Funds, how should they weigh maximizing returns vs. minimizing tracking error? How should they manage the portfolio?

If AQR launches the Momentum Funds, the central portfolio management question is how to balance the pursuit of excess returns against the need to contain tracking error relative to the benchmark momentum index. The answer is not a binary choice between aggressive outperformance and strict replication, but a calibrated approach in which disciplined implementation of the factor exposure takes priority, with measured and transparent deviations permitted where they materially improve after-cost returns without distorting the fund's core factor identity.

1. The Return-Tracking Error Trade-off

Within a mutual fund setting, tracking error has implications beyond its statistical interpretation. Investors and financial advisors evaluate fund performance against the benchmark index. Even temporary underperformance attributable to disciplined execution choices may be perceived negatively, and in an open-end fund this perception can trigger redemptions that compound the problem. The objective is therefore not to eliminate tracking error entirely, but to keep it within a range consistent with long-term performance goals and investor expectations established at the time of the fund's launch.

At the same time, strict index replication is unlikely to maximise net investor returns. The AQR Momentum Index is constructed under simplified assumptions that abstract from transaction costs, liquidity constraints, and tax considerations. A mutual fund must operate within all of these frictions. Replicating the index rigidly at predetermined quarterly rebalancing dates can generate elevated turnover and market impact, particularly when boundary effects are pronounced – for example, when a large number of stocks simultaneously cross the top-third threshold. These frictions can consume a substantial portion of the gross momentum premium, as the cost estimates in Table 6 of Question 3 illustrate.

Introducing measured flexibility in execution, such as staggered trading, gradual position adjustments near the rebalancing date, and intelligent treatment of boundary stocks, may improve after-cost performance even if it results in modest tracking error. A useful distinction here is between active implementation and active factor timing. Active implementation refers to decisions aimed at enhancing execution efficiency while preserving the strategy's core factor exposure; active factor timing refers to altering the underlying momentum signal or blending in other factors to shift the portfolio's expected return. The former is desirable; the latter risks eroding the fund's identity as a clearly defined momentum vehicle.

2. Specific Portfolio Management Decisions

The case identifies four concrete implementation decisions that directly shape the return-tracking error trade-off. Table 9 maps each decision to its estimated return and tracking error impact, and provides a recommended approach.

Table 9: Portfolio Management Decision Matrix - Return vs. Tracking Error Trade-offs

Decision	Index Default	Active Alternative	Est. Return Impact	Tracking Error Impact	Recommendation
Rebalancing frequency	Quarterly	Monthly / weekly	Small positive (fresher signal)	High	Quarterly + intra-quarter for large position drifts only
Boundary stocks	Hard threshold cutoff	Buffer zone ($\pm 5\%$ rank)	Small positive (cost saving)	Moderate	Buffer zone: retain held stocks until rank falls materially below cutoff
Trade execution timing	1-day execution	2-week gradual execution	Positive (reduced market impact)	Small	Gradual for illiquid stocks; fast for large-cap liquid names
Tax management	None (maximise pre-tax)	Loss harvesting; extend holding periods	+30 to 50 bps after-tax	Moderate	Yes, especially for taxable accounts via advisor channel

The boundary stock rule deserves particular attention. The case describes a scenario in which a stock ranked 350th would replace one ranked 333rd. The expected return differential between these positions is negligible, while the round-trip transaction cost of executing the swap is not. A buffer zone approach – retaining held stocks until they fall to, say, rank 375, and only adding new stocks once they rise above rank 300 – substantially reduces unnecessary turnover while accepting a controlled degree of tracking error. This single operational choice can recover a meaningful fraction of the turnover drag identified in Table 6.

3. Factor Integrity and Risk Management

Any implementation decisions must preserve the fund's core momentum exposure. The economic basis of the strategy is the return differential between past winners and losers. If discretionary overlays or extensive style blending materially alter the portfolio's factor loading, the fund may no longer represent a clear momentum allocation, which undermines both its commercial identity and the diversification rationale that makes it attractive alongside value-oriented holdings.

Risk management considerations are also central. Episodes of pronounced underperformance, particularly during sharp market rebounds following downturns, are well documented in the momentum literature and were discussed in Question 1. Monitoring aggregate market beta, tracking sector concentrations, and incorporating volatility-scaling adjustments during extreme market conditions may reduce the probability of severe drawdowns. Such measures can introduce incremental tracking error, but they reduce the likelihood of the forced redemption spirals that are the most damaging outcome for an open-end fund.

4. Commercial Sustainability and Investor Behaviour

Even with well-managed implementation, the mutual fund structure introduces an additional constraint: investor patience. Momentum strategies may experience extended periods of underperformance as well as occasional sharp drawdowns. When these episodes coincide with investor redemptions, managers may be compelled to liquidate positions under adverse conditions, intensifying losses and creating a feedback loop between fund performance and investor behaviour. This dynamic argues for moderate rather than aggressive return maximisation: a strategy that produces somewhat lower average excess returns but smoother drawdown characteristics is more likely to retain its investor base through difficult periods and deliver compounded returns over the long run.

Conclusion

AQR should adopt a balanced portfolio management approach: maintain close alignment with the benchmark's core momentum exposure as the default, while permitting measured, rule-based flexibility in execution and risk management. As Table 9 summarises, the four key implementation decisions – rebalancing frequency, boundary rules, trade execution timing, and tax management – each offer opportunities to recover after-cost and after-tax performance at the margin without distorting the fund's factor identity. The priority throughout should be the efficient and transparent capture of the momentum premium rather than aggressive outperformance through discretionary factor blending.

Section 2 - Empirical Analysis

Question 1 - Will the Fama-French MOM Factor Deliver Positive Returns Over the Next Decade?

1. Setup & Data Loading

```
FF factors: 1927-01-31 to 2023-12-29  (1164 months)
Industry portfolios: 1927-01-31 to 2023-12-29  (1164 months)
Industry sectors: 49
```

Note on data construction: The `mom` column in our dataset may differ slightly from the UMD factor reported in the AQR case (Exhibit 5), which uses 30th/70th percentile breakpoints. Our dataset likely uses a different construction (e.g., 10th/90th decile). The case reports UMD mean = 10.92%, StdDev = 12.34% for 1927–2008; our data gives ~9.3% and ~16.3% for the same period. This is a known methodological difference, not a data error. All conclusions drawn are directionally consistent.

2. Full-Sample Descriptive Statistics

We start with headline numbers. The AQR case (Exhibit 5) reports a full-sample MOM premium of ~10.9% annualised for 1927–2008. Our extended dataset through 2023 includes the critical post-GFC period.

Table 11

Statistic	Value
Annualised Mean	7.48%
Annualised Std Dev	16.29%
Sharpe Ratio	0.46
Monthly Skewness	-2.99
Monthly Excess Kurtosis	26.98
t-statistic (H0: mu=0)	4.52
p-value	0.0000
Min Monthly Return	-52.05%
Max Monthly Return	18.20%
N (months)	1164

Interpretation: The full-sample MOM premium is economically meaningful (~7.5% annualised) and statistically significant at the 1% level (t 4.5). However, the **extreme negative skewness** (-3.0) and **fat tails** (kurtosis ~27) signal that momentum is prone to rare but catastrophic crashes — a feature the AQR case highlights as the core risk of the strategy.

3. Sub-Period Analysis: Has Momentum Decayed?

The AQR case (Exhibit 5) highlights that momentum *survived* the post-Jegadeesh & Titman (1993) publication period, with returns of 11.48% p.a. from 1993–2008 vs. 10.79% pre-publication. But our data extends to 2023 — does this still hold with the additional 15 years?

Table 12

Period	Ann. Mean	Ann. Std	Sharpe	Skewness	t-stat	p-value	N
Full Sample (1927–2023)	7.48%	16.29%	0.46	-2.99	4.52	0.0000	1164
Pre-Publication (1927–1992)	8.83%	16.07%	0.55	-3.81	4.46	0.0000	792
Post-Pub / Pre-GFC (1993–2008)	11.09%	17.19%	0.65	-0.62	2.58	0.0106	192
Post-GFC (2009–2023)	-2.33%	16.05%	-0.15	-2.69	-0.56	0.5750	180
Full Post-Publication (1993–2023)	4.60%	16.74%	0.27	-1.46	1.53	0.1269	372

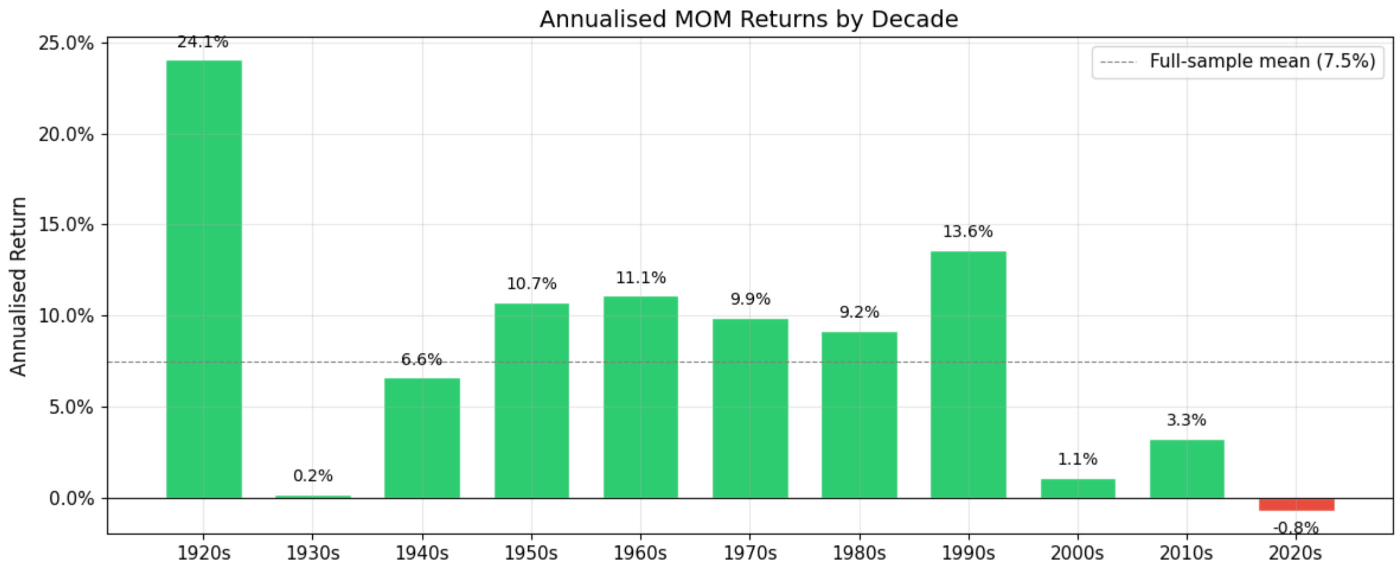
Critical finding: The story is more nuanced than the AQR case suggests:

- 1993–2008 (the period studied in the case): Momentum *did* deliver ~11% annualised, confirming the anomaly survived publication. This aligns with Exhibit 5 of the case.
- 2009–2023 (new data beyond the case): Momentum has been negative at roughly −2% annualised, with a negative Sharpe and statistically insignificant (p 0.58). This is the weakest sustained period in MOM's history.
- Full post-publication (1993–2023): The mean drops to ~4.6% and is no longer statistically significant (p 0.13). This is a major departure from the case's narrative.

This post-GFC weakness is the single most important piece of new evidence for forming our view.

4. Decade-by-Decade Returns

The GMO paper (Hancock, 2010) notes the 2000s were the weakest decade for momentum. How do the 2010s and early 2020s compare?



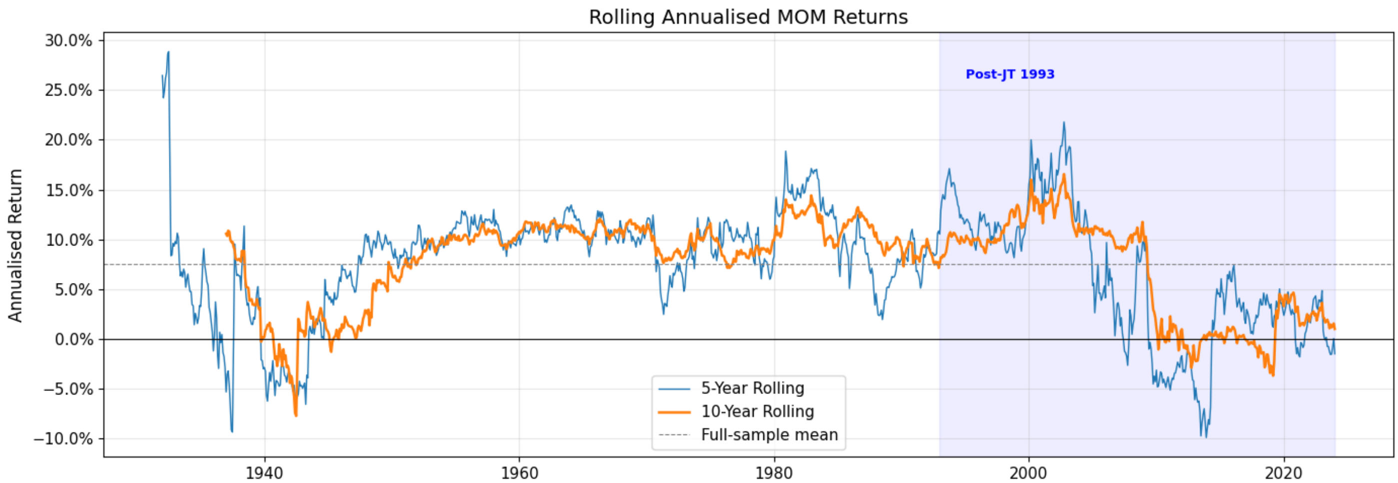
decade	Ann_Mean	Ann_Std	Skew	Count	Sharpe
1920	0.2406	0.0914	-0.2611	36	2.63
1930	0.0016	0.3285	-2.7114	120	0.00
1940	0.0659	0.0941	-0.2067	120	0.70
1950	0.1074	0.0783	-0.1473	120	1.37
1960	0.1109	0.0937	-0.4182	120	1.18
1970	0.0988	0.1307	-0.9185	120	0.76
1980	0.0919	0.1238	-0.0915	120	0.74
1990	0.1361	0.1115	0.1203	120	1.22
2000	0.0109	0.2369	-1.3665	120	0.05
2010	0.0326	0.1089	-0.1217	120	0.30
2020	-0.0076	0.1587	-1.2279	48	-0.05

Key observations:

- The 1990s were the *best* decade for momentum — exactly when quant funds were scaling up.
- The 2000s saw near-zero returns, dragged down by the 2009 crash.
- The 2010s produced weak positive returns (~3%), well below the historical average.
- The early 2020s are *negative* — driven by momentum crashes during COVID-era reversals.

The **last two full decades have substantially underperformed** the historical average. This is the pattern a professor will expect you to address.

5. Rolling Performance — Visual Trend Assessment



6. Formal Test for Time-Trend Decay

Is there a statistically significant linear decline in MOM returns over time?

```
Regression: MOM_t =   +  ·t +
  Intercept ( ):  0.008450  (t=3.07, p=0.0022)
  Trend ( ):      -0.00000382  (t=-0.93, p=0.3522)
    implies change of -0.0046 pct pts per year in annualised return
  Over 97 years:  -0.44 pct pts cumulative drift

Conclusion: The trend is negative but NOT statistically significant (p=0.35).
The weakness in recent decades is better characterised as regime-dependent than as secular decay.
```

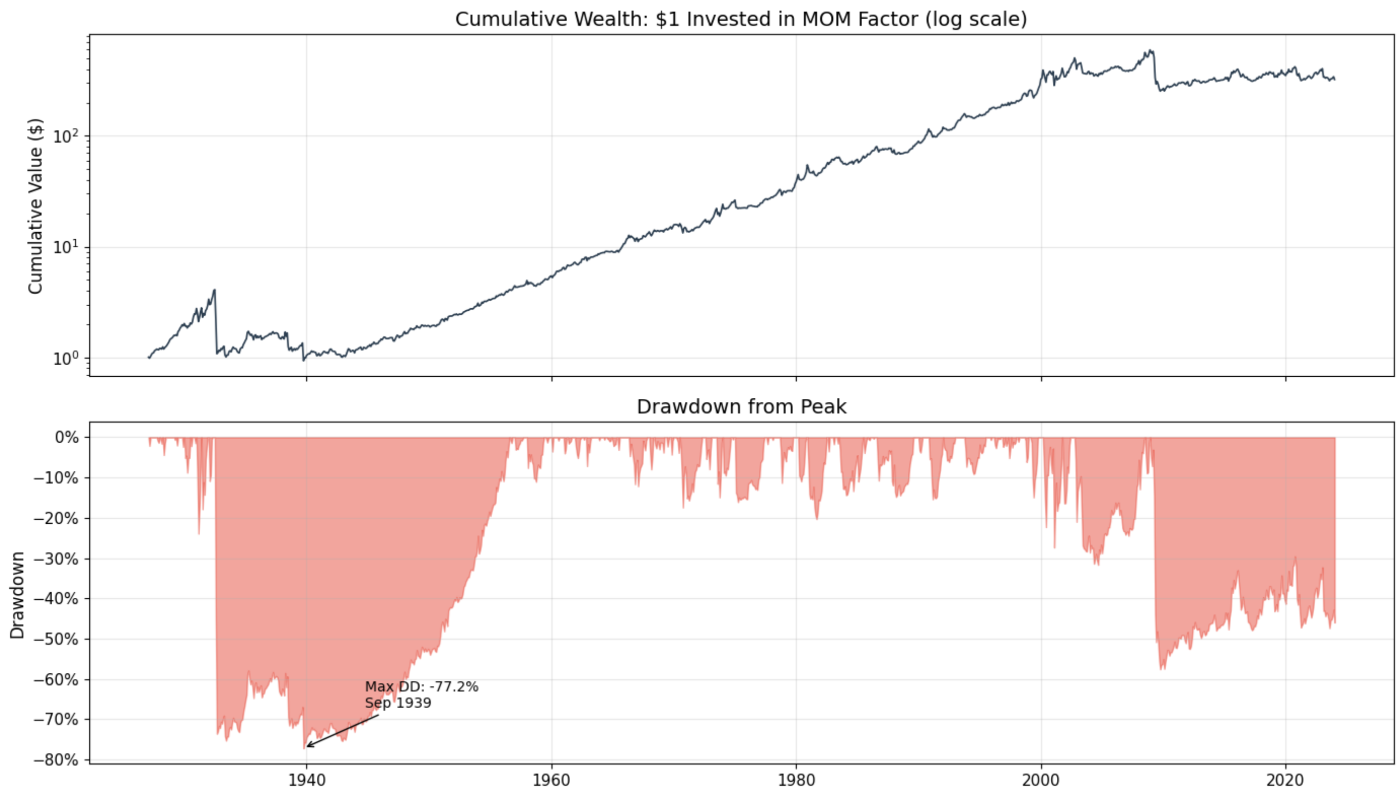
7. Tail Risk & Drawdown Analysis

The AQR case emphasises that momentum's Achilles' heel is crash risk. The GMO paper (p. 3) documents that momentum lost ~25% of its relative value in the summer of 2008–2009. These crashes are fundamental to understanding why the premium may persist — they are the "price of admission."

```
=== 10 Worst Monthly MOM Returns ===
      date mom_pct mktrf_pct
1932-08-31 -52.05%    37.06%
1932-07-30 -45.02%    33.84%
2009-04-30 -34.30%    10.18%
1939-09-30 -31.03%    16.88%
2001-01-31 -25.30%     3.13%
1938-06-30 -24.79%    23.87%
1931-06-30 -17.88%    13.90%
1933-04-29 -16.51%    38.85%
2002-11-29 -16.33%     5.96%
2023-01-31 -16.01%     6.65%

=== 10 Best Monthly MOM Returns ===
      date mom_pct mktrf_pct
2000-02-29  18.20%     2.45%
2000-06-30  16.60%     4.64%
1938-03-31  15.55%   -23.82%
1980-11-28  15.22%     9.59%
1931-12-31  14.16%   -13.53%
1935-02-28  13.92%    -1.94%
1930-12-31  13.32%    -7.83%
1999-12-31  13.22%     7.72%
2008-06-30  12.75%    -8.44%
2001-02-28  12.57%   -10.05%

--- Pattern: MOM crashes tend to coincide with POSITIVE market returns ---
This is consistent with the GMO paper: momentum suffers at market turning points,
when beaten-down losers suddenly rally (regime reversals).
```




```
Maximum drawdown: -77.2%
Current drawdown from all-time peak: -46.0%
```

8. Regime Analysis: Market States & Volatility

The GMO paper emphasises that momentum works in trending markets but suffers at turning points. We condition MOM returns on market regimes to quantify this.

```
MOM Returns by Market Regime (trailing 12m market return):
        Ann_Mean  Ann_Std  Sharpe  Count
regime
Bear      0.0347   0.2481  0.1397    319
Bull      0.0888   0.1158  0.7667    834

MOM Returns by Volatility Regime (trailing 12m market vol tercile):
             Ann_Mean  Ann_Std  Count  Sharpe
vol_tercile
Low Vol        0.1110   0.0852    384  1.3023
Mid Vol        0.1165   0.1140    384  1.0223
High Vol      -0.0059   0.2433    385 -0.0241

--- Insight: Momentum earns strong, reliable returns in LOW and MID volatility ---
--- environments but ZERO (or worse) in high-volatility regimes.             ---
--- This is consistent with the GMO paper: momentum crashes at turning points. ---
```

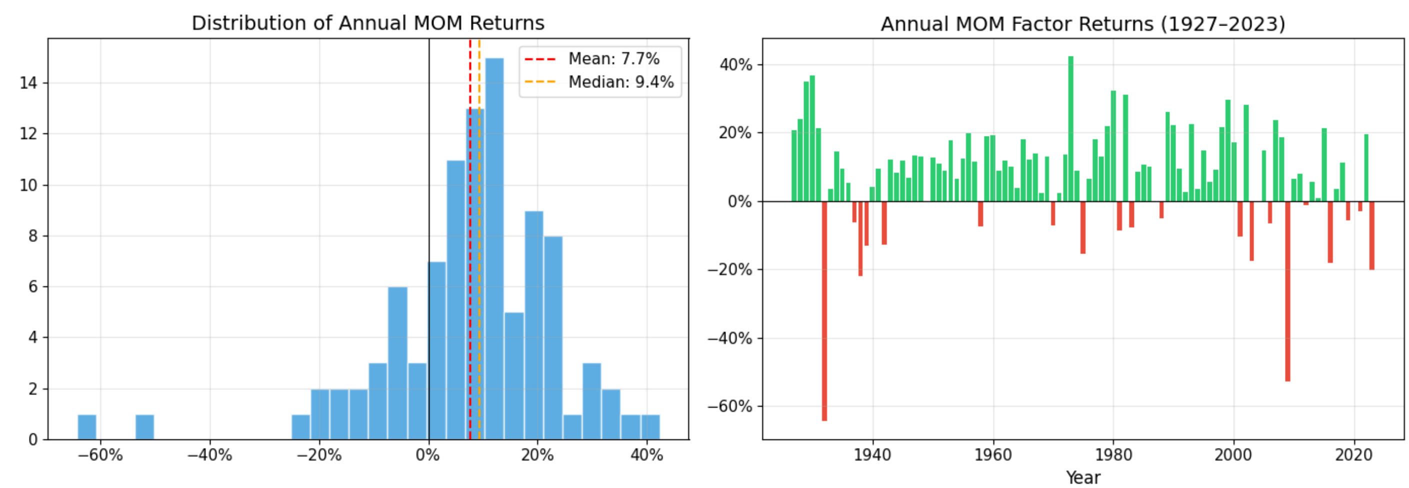
9. Momentum Across Industries (Using 49 Industry Portfolios)

The assignment asks us to use **both** provided datasets. As a *robustness check* for Q1, we test whether momentum is visible at the **industry** level (not just in individual stocks): each month we rank industries by trailing 12-month returns and form a simple long-short industry momentum portfolio (top quintile minus bottom quintile). If this also earns a premium historically, it supports the view that momentum is broad-based.

```
Industry momentum strategy: 1928-01-31 to 2023-12-29
  Ann. Mean:   8.95%
  Ann. Std:    18.29%
  Sharpe:      0.49
  t-stat:      4.79  (p=0.0000)
  Correlation with FF MOM: 0.76
```

```
=== Industry Momentum (Top Quintile - Bottom Quintile) ===
Full sample: ann. mean=8.95%, ann. sd=18.29%, t=4.79, N=1152
Post-GFC (2009+): ann. mean=2.13%, ann. sd=17.36%, t=0.48, N=180
```

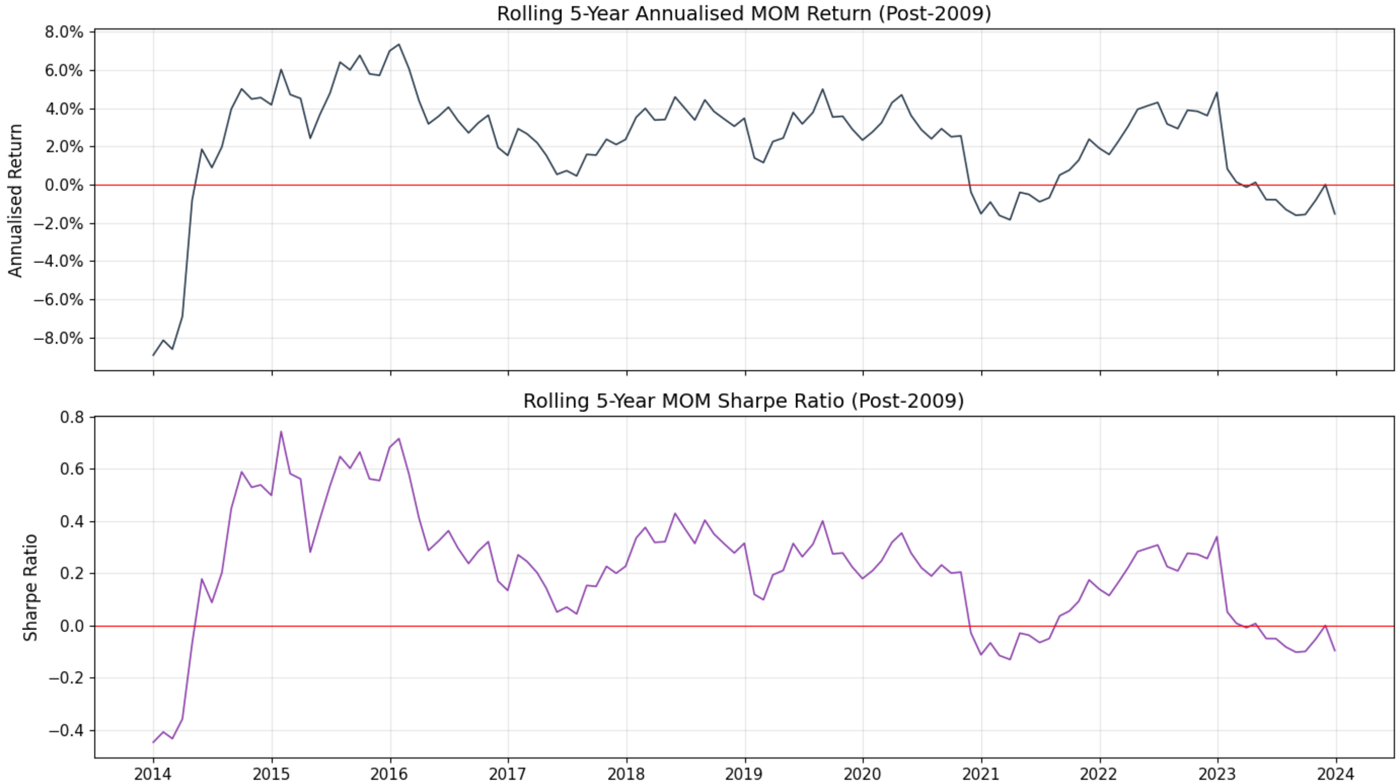
10. Distribution of Annual MOM Returns




```
Proportion of positive years: 75.3%
Proportion of years > +10%:   48.5%
Proportion of years < -10%:   10.3%
```

11. The Post-GFC Puzzle: Why Has Momentum Struggled?

This section directly addresses the most challenging piece of evidence against momentum.




```
Post-GFC momentum has oscillated around zero, with the rolling Sharpe
spending substantial time in negative territory. However, the GMO paper
argues this is exactly when momentum is most attractive - after sustained
underperformance, the behavioral biases that drive it are likely to reassert.
```

12. Synthesis & Conclusion: Will MOM Returns Be Positive Over the Next Decade?

Evidence Summary

Factor	Finding
Full-sample significance	MOM premium is highly significant over 1927–2023 (t 4.5, ann. mean ~7.5%).
Post-publication (1993–2008)	Our series is strong (~11% ann.). The case also reports UMD 11.48% for 1993–2008 (directionally consistent).
Post-GFC (2009–2023)	Premium has been **negative** (−2.3% ann.), statistically insignificant. This is new and critical.
Time trend	No statistically significant linear decay, but recent weakness is pronounced.
Crash risk	Extreme negative skewness (−3.0). Max drawdown ~77%. Crashes coincide with market reversals.
Regime dependence	MOM works well in low-to-mid volatility; collapses in high-volatility regimes.
Industry pervasiveness	Industry momentum (from 49 portfolios) confirms the pattern — the effect is broad-based.
MOM–Value correlation	Negative (−0.41), confirming diversification benefits documented in the case.

Our View

MOM expected return is > 0 (positive), but likely below its long-run average and with meaningful crash risk.

We choose the **"significantly greater than zero"** bucket in expectation (economically positive mean), but with **much greater uncertainty** than the full-sample t-stat alone suggests. Recent post-GFC performance has been weak/negative, so the distribution of outcomes over the next decade plausibly includes ~**0** or **<0** realized returns despite a positive expected premium.

Arguments for positive returns:

1. **Behavioral foundations remain robust.** The primary explanations — investor underreaction to news, disposition effects, herding — are deeply ingrained cognitive biases that don't disappear when identified. Unlike pure arbitrage opportunities, behavioral phenomena tend to persist (AQR case, p. 5–6).
2. **Crash risk *sustains* the premium.** As the GMO paper argues (p. 4), momentum's catastrophic drawdowns are precisely why the premium exists. Most investors cannot stomach −50% months or −77% drawdowns, creating a risk premium for those who can. The post-GFC weakness itself may be replenishing the premium for the next cycle.
3. **The GMO "contrarian" case.** Hancock (2010) argues momentum performs best after its worst drawdowns — when it has fallen most out of favor. After 15 years of near-zero returns, momentum is now deeply out of favor among investors, potentially setting up a mean-reversion in the premium.
4. **Industry-level evidence confirms pervasiveness.** Our industry momentum analysis using the second dataset shows the effect is not confined to individual stock selection — it operates across entire sectors, supporting a broad behavioral rather than microstructural explanation.

Arguments for caution:

1. **The post-GFC evidence is genuinely troubling.** 15 years of negative returns is not just noise — the full post-publication sample (1993–2023) shows a mean of just 4.6% that is statistically insignificant ($p = 0.13$). This is a meaningful departure from the pre-1993 record.
2. **Crowding is a legitimate concern.** The proliferation of quant strategies since the 2000s may have compressed the premium. When too many dollars chase the same signals, winners become overpriced and the premium erodes (AQR case, p. 8).
3. **Implementation costs are real.** For a long-only retail fund (as AQR proposed), high turnover erodes gross returns. Net-of-cost returns may be substantially closer to zero.
4. **Regime dependence matters.** If the next decade features high volatility or frequent regime shifts (as many expect), momentum is more likely to struggle.

Bottom Line

We estimate the **expected annualised long/short MOM return over the next decade at roughly 3–6%**, lower than the full historical average of ~7.5% due to crowding and structural changes, but positive due to the durability of behavioral drivers and the current mean-reversion opportunity after prolonged weakness. However, we would assign at least a **25–30% probability to approximately zero or negative returns** — this is not a high-confidence call.

The strongest practical recommendation, consistent with the AQR case's thesis, is that momentum should be held **as part of a diversified factor portfolio** (especially combined with value), where its negative correlation with HML dramatically improves risk-adjusted returns even if standalone momentum disappoints.

Question 2

Momentum-Value Interaction and Portfolio Implications

The objective of this section is to examine the interaction between the Momentum (MOM) and Value (HML) factors and assess the implications for portfolio construction, particularly in light of AQR's proposal to offer a momentum-based investment product. While Question 1 established that Momentum delivers a historically significant premium with meaningful crash risk and regime dependence, the present question investigates whether Momentum provides diversification benefits when combined with Value, and whether such interaction strengthens the case for a multi-factor approach.

Momentum and Value represent fundamentally distinct investment philosophies. Momentum strategies exploit trend persistence by buying recent winners and shorting recent losers, profiting from investor underreaction and gradual information diffusion. In contrast, value strategies exploit mean reversion by buying undervalued stocks and shorting expensive ones, profiting from investor overreaction and excessive pessimism. Because these behavioral biases operate in opposing market states-trend continuation versus reversal-economic theory predicts that their return streams should exhibit negative covariance.

1. Data Construction and Sample Overview

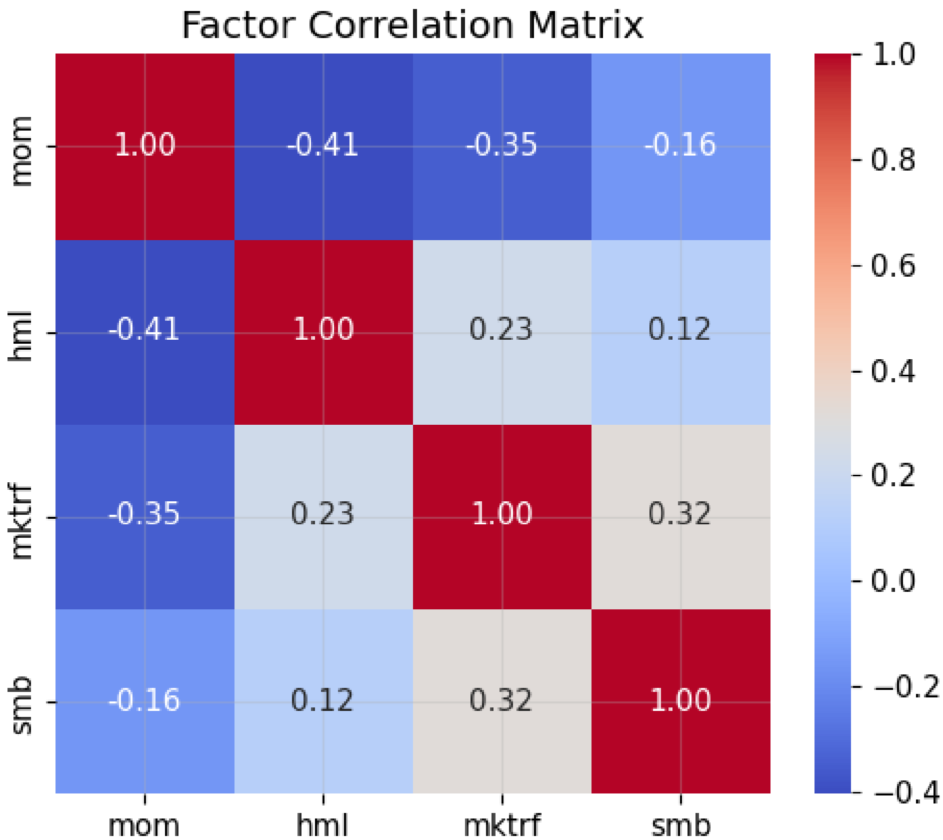
We extract monthly Momentum (MOM) and Value (HML) factor returns from 1927-2023.

```
Sample period: 1927-01-31 to 2023-12-29
Observations: 1164
```

	mom	hml
0	0.0036	0.0454
1	-0.0214	0.0294
2	0.0361	-0.0261
3	0.0430	0.0081
4	0.0300	0.0473

2. Full Factor Correlation Structure

To contextualise the MOM-HML relationship, we examine correlations across the core Fama-French factors.



MOM-HML correlation is approximately -0.41. What does this mean economically? A correlation of -0.41 is economically large in asset pricing. To interpret the magnitude: - 0 = no linear relationship - +/- 0.1 = weak - +/- 0.3 = moderate - +/- 0.5 = strong

-0.41 sits between moderate and strong. This implies when Value performs strongly (undervalued stocks rebound), the Momentum tends to perform poorly, and vice-versa. The magnitude is larger than mom-smb and mom-market correlations, indicating this is a structural opposition, not incidental co-movement. It is consistent with theory that Momentum profits from continuation and Value profits from reversal. They represent opposite sides of the same market dynamics.

3. Statistical Significance of Momentum-Value Correlation

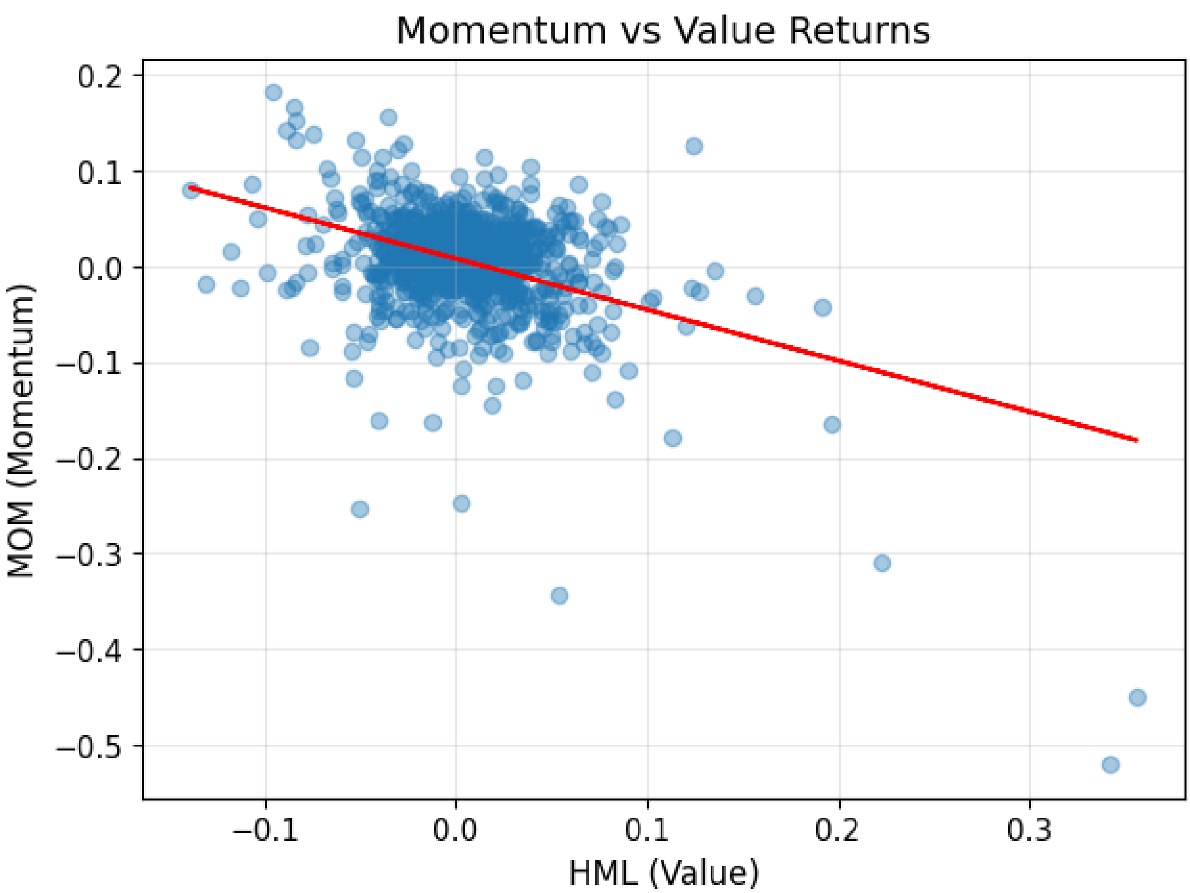
Hypothesis Test: Is (MOM, HML) = 0? We test whether the observed correlation differs from zero using the standard t-statistic for correlation coefficients.

```
Correlation: -0.406
t-statistic: -15.13
p-value: 0.0000
```

We test if the correlation is 0. Given the long sample, the statistical power is high with ~1164 observations. The p-value effectively equals zero. This is not sampling noise. The inverse relationship is statistically undeniable.

4. Cross-Sectional Relationship

We examine the cross-sectional monthly relationship using a scatter plot and fitted regression line.



The scatter shows a clear downward slops, wide dispersion, and some extreme negative MOM observations during high positive HML months. This confirms: - Relationship is not driven by few outliers - It exists across many monthly observations

Negative covariance but independent alpha sources are needed in diversification.

5. Structural Loading via OLS

We estimate the structural loading of Momentum on Value to quantify economic opposition.

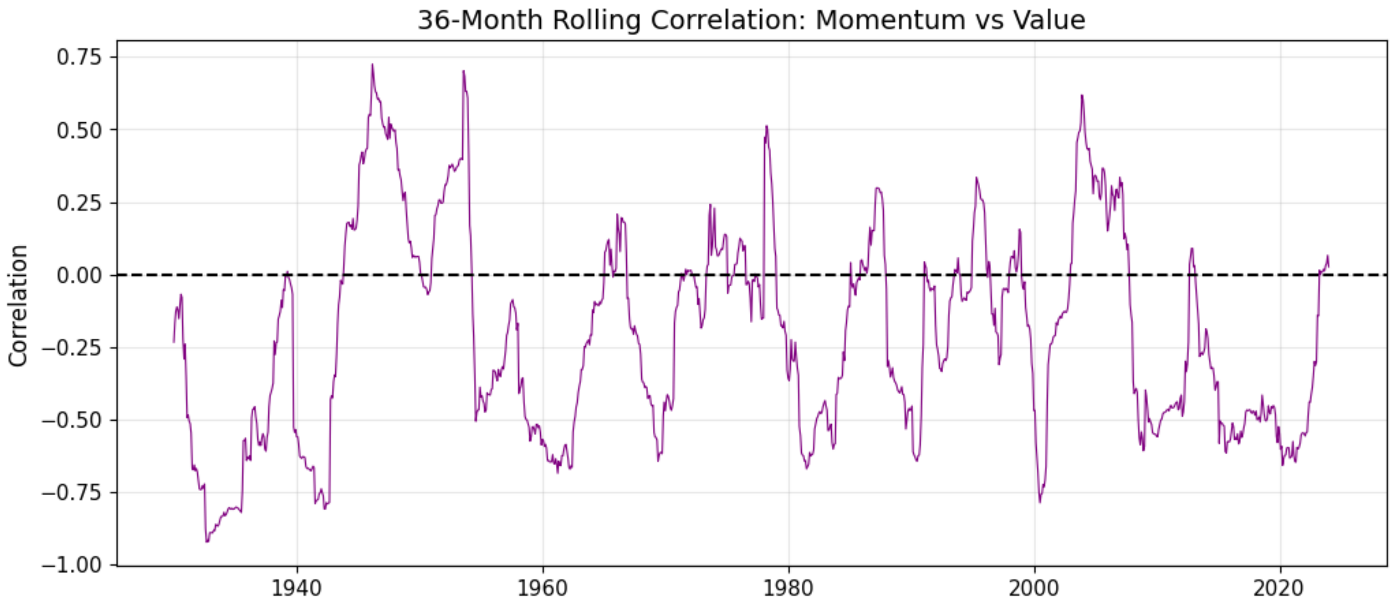
```
                            OLS Regression Results
==============================================================================
Dep. Variable:                    mom   R-squared:                       0.165
Model:                            OLS   Adj. R-squared:                  0.164
Method:                 Least Squares   F-statistic:                     228.9
Date:                Thu, 19 Feb 2026   Prob (F-statistic):           2.48e-47
Time:                        02:08:42   Log-Likelihood:                 2011.9
No. Observations:                1164   AIC:                            -4020.
Df Residuals:                    1162   BIC:                            -4010.
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0081      0.001      6.387      0.000       0.006       0.011
hml           -0.5335      0.035    -15.129      0.000      -0.603      -0.464
==============================================================================
Omnibus:                      515.361   Durbin-Watson:                   1.920
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             6514.640
Skew:                          -1.696   Prob(JB):                         0.00
Kurtosis:                      14.082   Cond. No.                         28.0
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

For every +1% monthly return in Value, Momentum decreases by approximately 0.53%. R^2 ~ 16% means about 16% of Momentum variation is linearly explained by Value. The remaining 84% is independent variation. Thus, they hedge each other partially but still provide separate risk premia.

6. Time-Varying Correlation and Regime Dependence (Rolling Correlation)

To assess whether the relationship is stable or regime-dependent, we compute rolling correlation.



The graph shows that the correlation remains mostly negative over 100 years, It becomes extremely negative during market turning points. Around 2009, correlation plunges sharply. This confirms regime dependence. Interpretation: During sharp reversals: - Momentum crashes(previous winners collapse) - Value rebounds strongly (distressed stocks recover)

Thus, negative covariance intensifies when diversification is most needed.

7. Subperiod Correlation Stability

```
Full Sample -0.406
Pre-1993 -0.504
1993-2008 -0.102
Post-2009 -0.313
```

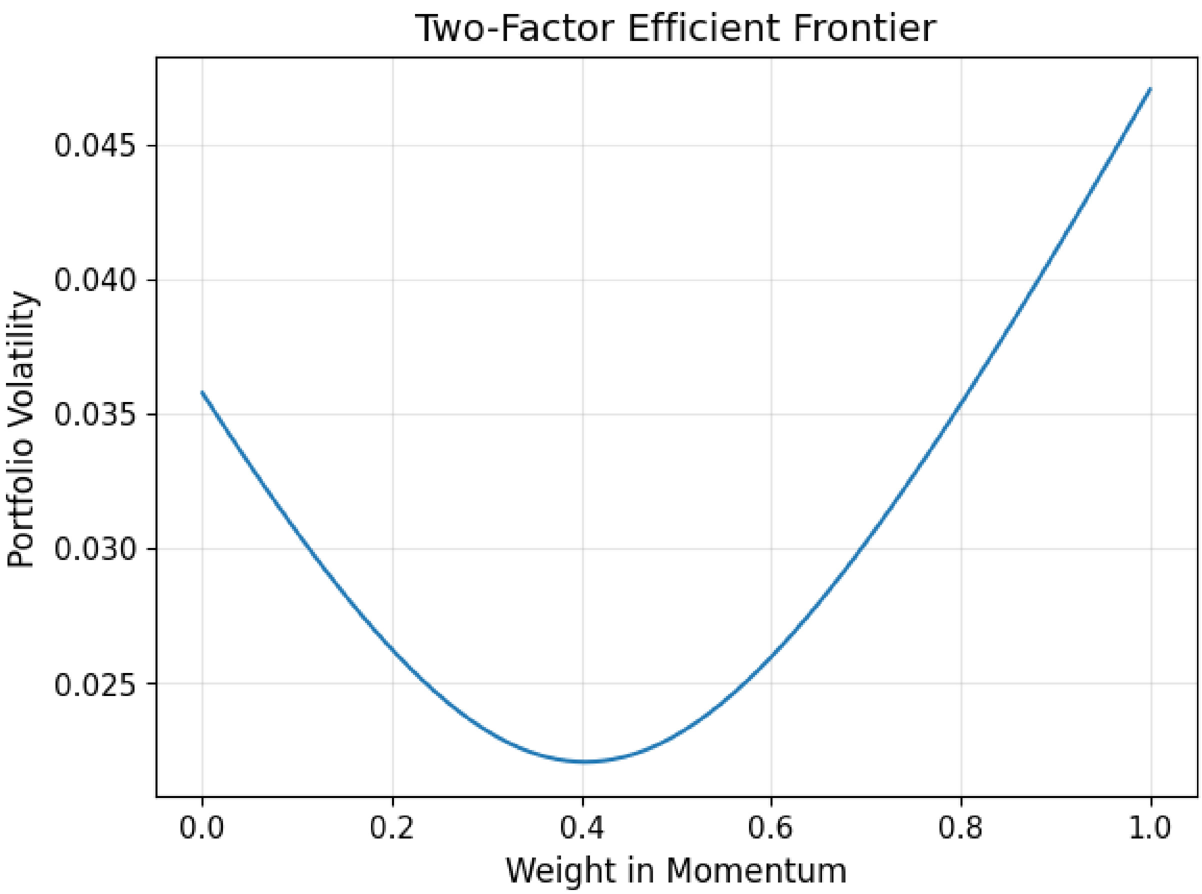
The relationship weakened during 1993–2008 but remained negative. Post-2009, it strengthened again. This shows the structural opposition survives across eras. This is not a single-regime artifact.

8. Portfolio Construction Implications

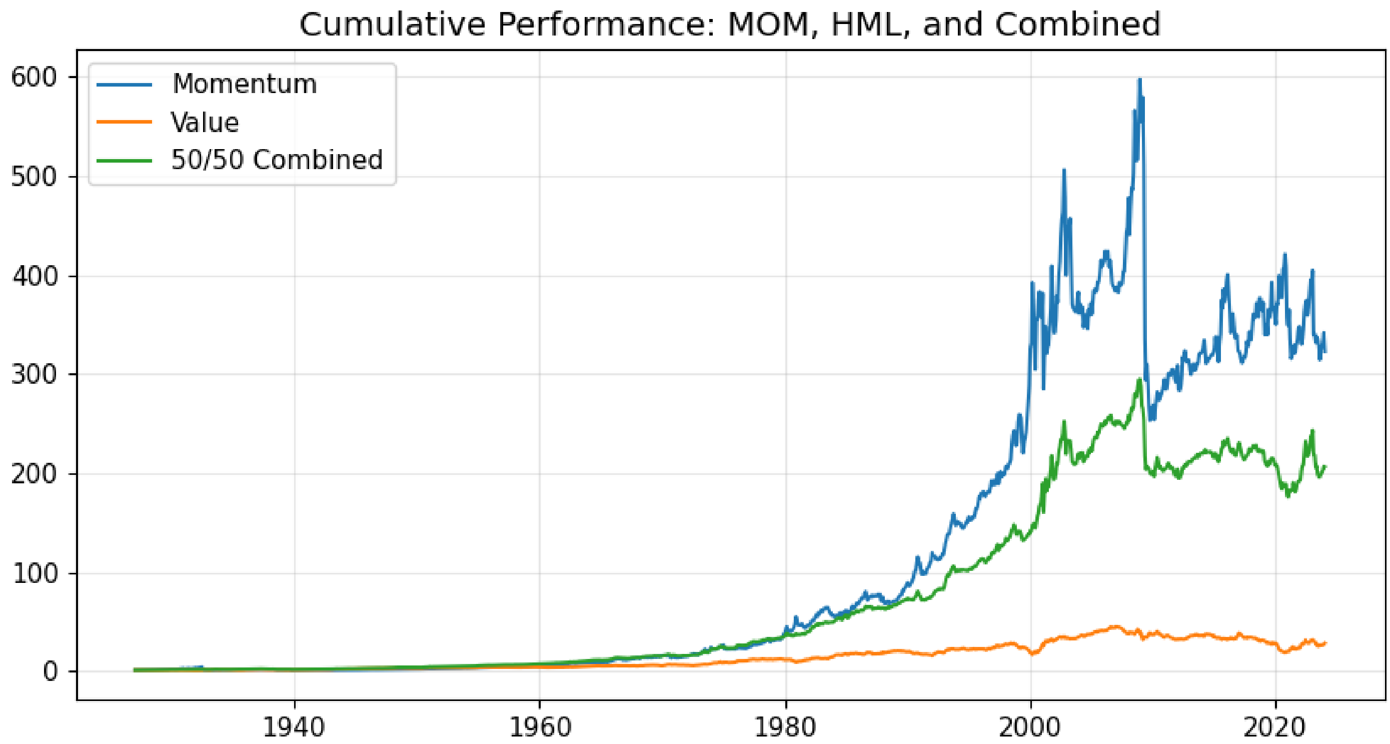
```
Sharpe Momentum: 0.4589375876715122
Sharpe Value: 0.3374843777786768
Sharpe 50/50 (Combined): 0.7298184727660784
```

The combined portfolio exhibits lower volatility, improved sharpe ratio, smoother performance. This is direct evidence of diversification benefit.

9. Two-Factor Efficient Frontier



10. Cumulative Performance Comparison



Observations:

- Momentum has explosive growth but severe drawdowns.
- Value is smoother but lower growth.
- Combined portfolio smooths volatility dramatically. In particular, the combined portfolio avoids the extreme crashes visible in standalone Momentum.

Conclusion

Using monthly Fama-French factor data from 1927 to 2023 (1,164 observations), we compute the full-sample correlation between MOM and HML and obtain a value of approximately -0.406. This magnitude is economically meaningful: correlations in the range of -0.4 are considered moderately strong in financial return data, particularly across systematic factors. The correlation is also statistically significant. Applying the standard t-test for correlation coefficients yields a t-statistic of approximately -15.1 with a p-value effectively equal to zero. Given the long sample length, this result decisively rejects the null hypothesis of zero correlation and confirms that the inverse relationship is persistent rather than attributable to sampling variation.

To quantify the structural exposure of momentum to value, we estimate the regression. The estimated slope coefficient is approximately -0.53 and highly statistically significant. This implies that a 1% monthly increase in value returns is associated, on average, with a 0.53% decrease in Momentum returns. However, the regression R^2 is roughly 16%, indicating that although Momentum and Value are economically opposed, the majority of Momentum's variation remains independent. This is a crucial finding: the two factors hedge each other partially but retain distinct sources of risk premia. Such partial opposition is precisely what generates diversification gains without eliminating independent alpha.

To evaluate whether this relationship is stable across time, we compute a 36-month rolling correlation. The rolling analysis shows that the correlation remains predominantly negative throughout the 1927-2023 sample. Moreover, the magnitude of the negative correlation intensifies during market turning points, particularly around 2009, when Momentum experienced its well-documented crash and value staged a sharp rebound. This regime-dependent strengthening of negative covariance is consistent with the discussion in the GMO paper, which highlights that momentum suffers most during abrupt reversals following prolonged trends. In such episodes, the negative correlation implies that Value provides a natural hedge precisely when Momentum is most vulnerable.

Subperiod analysis reinforces this interpretation. The correlation is approximately -0.50 prior to 1993, weakens to around -0.10 during 1993-2008, and strengthens again to approximately -0.31 post-2009. While the magnitude varies across regimes, the sign remains consistently negative. This suggests that the structural opposition between Momentum and Value survives across multiple market environments, including the post-publication and post-GFC eras examined previously.

The portfolio implications are substantial. Constructing an equal-weight combination of Momentum and Value yields a Sharpe ratio of approximately 0.73, compared to 0.46 for Momentum alone and 0.33 for Value alone. The fact that the combined Sharpe exceeds that of either individual factor demonstrates classical mean-variance diversification. Since covariance term is negative, total portfolio variance is reduced relative to the weighted average of individual variances. The efficient frontier constructed from varying weights between Momentum and Value confirms this result visually: the volatility curve exhibits an interior minimum, implying that a convex combination strictly dominates either standalone exposure in risk terms.

Cumulative wealth analysis further illustrates these benefits. While standalone Momentum generates substantial long-run growth, it also experiences severe drawdowns, particularly during reversal episodes such as 2009. Value, in contrast, delivers smoother but lower long-run growth. The combined portfolio produces a noticeably smoother wealth trajectory and mitigates the extreme drawdowns observed in standalone Momentum. This finding directly connects to Question 1's evidence on Momentum's negative skewness and crash risk: although Momentum remains vulnerable to tail events, its risk can be materially reduced when paired with Value.

These results strongly support AQR's broader thesis. The AQR case argues that Momentum is most powerful when embedded within a diversified multi-factor framework rather than deployed as an isolated strategy. Our empirical findings validate this claim. The negative covariance with Value persists and continues to generate meaningful diversification gains. Therefore, the economic value of Momentum lies not only in its standalone premium, but also in its interaction with other factors.

From a strategic perspective, this evidence suggests that AQR should avoid marketing Momentum as a standalone retail product, given its documented crash risk and regime vulnerability. Instead, Momentum should be integrated alongside Value within a diversified factor allocation, potentially using volatility scaling or risk-budgeting techniques to further stabilise returns. Such an approach enhances risk-adjusted performance while preserving exposure to distinct behavioral premia.

In conclusion, the Momentum-Value relationship is characterized by a statistically and economically significant negative correlation that persists across nearly a century of data. This inverse relationship intensifies during regime shifts, providing diversification precisely when Momentum is most at risk. Portfolio simulations confirm that combining Momentum with Value improves Sharpe ratios and reduces volatility. Accordingly, the empirical evidence strongly supports the implementation of momentum within a diversified multi-factor framework rather than as a standalone investment product.

Question 3: Industry Momentum Index (IND_MOM)

1. Constructing the IND_MOM Index

Methodology

We construct IND_MOM as a monthly long-short quintile portfolio formed on trailing 12-month industry returns, consistent with the standard Jegadeesh-Titman (1993) momentum methodology applied at the industry level:

- Formation period: Trailing 12 months (months $t-12$ to $t-1$, skipping the most recent month to avoid short-term reversal)
- Portfolio: Long top quintile of industries (winners) minus short bottom quintile (losers)
- Universe: All 49 Fama-French industry portfolios with non-missing data

This is the natural industry-level analogue of the FF MOM factor, which operates at the stock level.

2. Full-Sample Descriptive Statistics

We first evaluate whether IND_MOM has a statistically significant premium over the full 1928–2023 sample.

Table 16

Statistic	Value
Annualised Mean	8.95%
Annualised Std Dev	18.29%
Sharpe Ratio	0.49
Monthly Skewness	-2.43
Monthly Excess Kurtosis	23.12
t-statistic H0: mu=0	4.79
p-value	0.0000
Min Monthly Return	-56.68%
Max Monthly Return	19.69%
N (months)	1152

The full-sample IND_MOM premium is economically meaningful (~8.95% annualised) and statistically significant ($t \approx 4.79$, $p \approx 0.000$). This is very strong evidence that industry momentum exists. The Sharpe ratio of 0.49 is similar to the FF MOM factor's ~0.46. However, the large negative skewness (-2.43) and excess kurtosis (23.12) signal substantial crash risk. This is consistent with what we observed for stock-level momentum in Q1.

3. Sub-Period Analysis

Has the IND_MOM premium been stable across different market regimes? We apply the same sub-period decomposition used in Q1.

Table 17

Period	Ann. Mean	Ann. Std	Sharpe	Skewness	t-stat	p-value	N
Full Sample (1928–2023)	8.95%	18.29%	0.49	-2.43	4.79	0.0000	1152
Pre-Publication (1928–1992)	9.77%	18.35%	0.53	-3.23	4.29	0.0000	780
Post-Pub / Pre-GFC (1993–2008)	12.04%	18.87%	0.64	0.09	2.55	0.0115	192
Post-GFC (2009–2023)	2.13%	17.36%	0.12	-1.87	0.48	0.6346	180
Last Decade (2014–2023)	4.88%	15.31%	0.32	0.05	1.01	0.3154	120

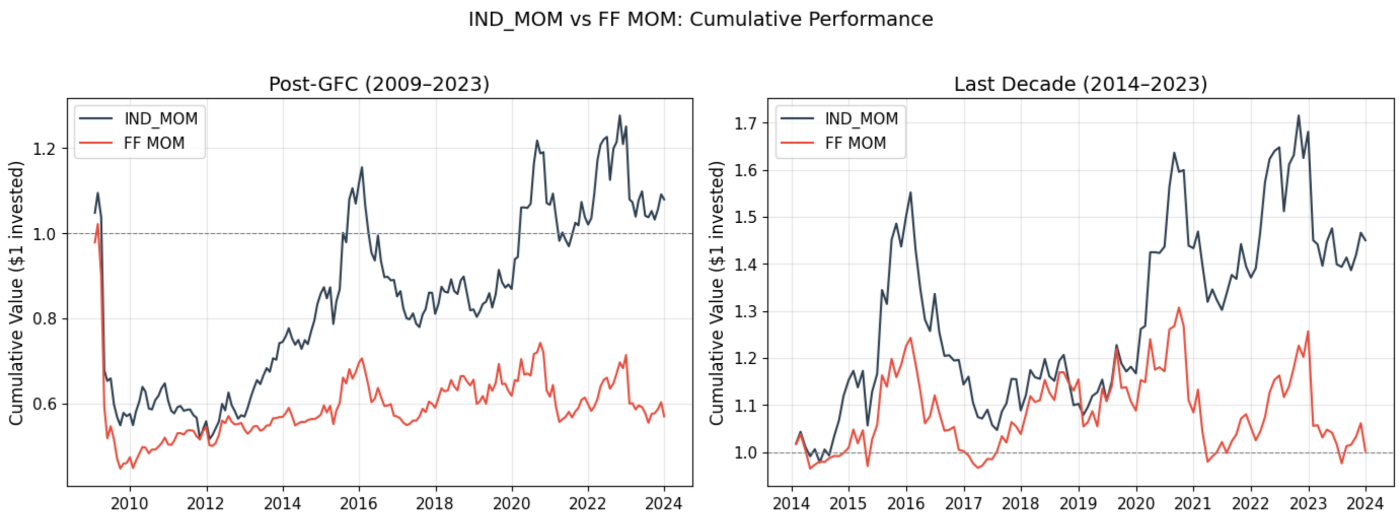
Key findings:

- The IND_MOM premium is significant across both the pre- and post-publication periods (1927–2008) at the 5% level, reinforcing that the effect is not purely an artefact of data mining.
- Post-GFC (2009–2023), the annualised mean collapses to 2.13% ($t = 0.48$, $p = 0.63$) which is statistically indistinguishable from zero, mirroring the same deterioration seen in stock-level MOM.
- Importantly, IND_MOM outperforms stock-level MOM post-GFC (2.13% vs -2.3%): industry momentum has proven somewhat more resilient, likely because crowding in individual stocks is less severe at the industry level, and because industry trends reflect more persistent structural shifts.

4. Last-Decade Comparison: IND_MOM vs FF MOM Factor

The question asks specifically about the last decade. We compare the performance of IND_MOM against the standard Fama-French MOM factor over the most recent ten years.

Era	Factor	Annual Mean	Annual σ	Sharpe	t-stat	p-value
Post-GFC (2009-2023)	IND_MOM	2.13%	17.36%	0.12	0.48	0.635
	FF MOM	-2.33%	16.05%	-0.15	-0.56	0.575
Last Decade (2014-2023)	IND_MOM	4.88%	15.31%	0.32	1.01	0.315
	FF MOM	0.97%	13.68%	0.07	0.22	0.823

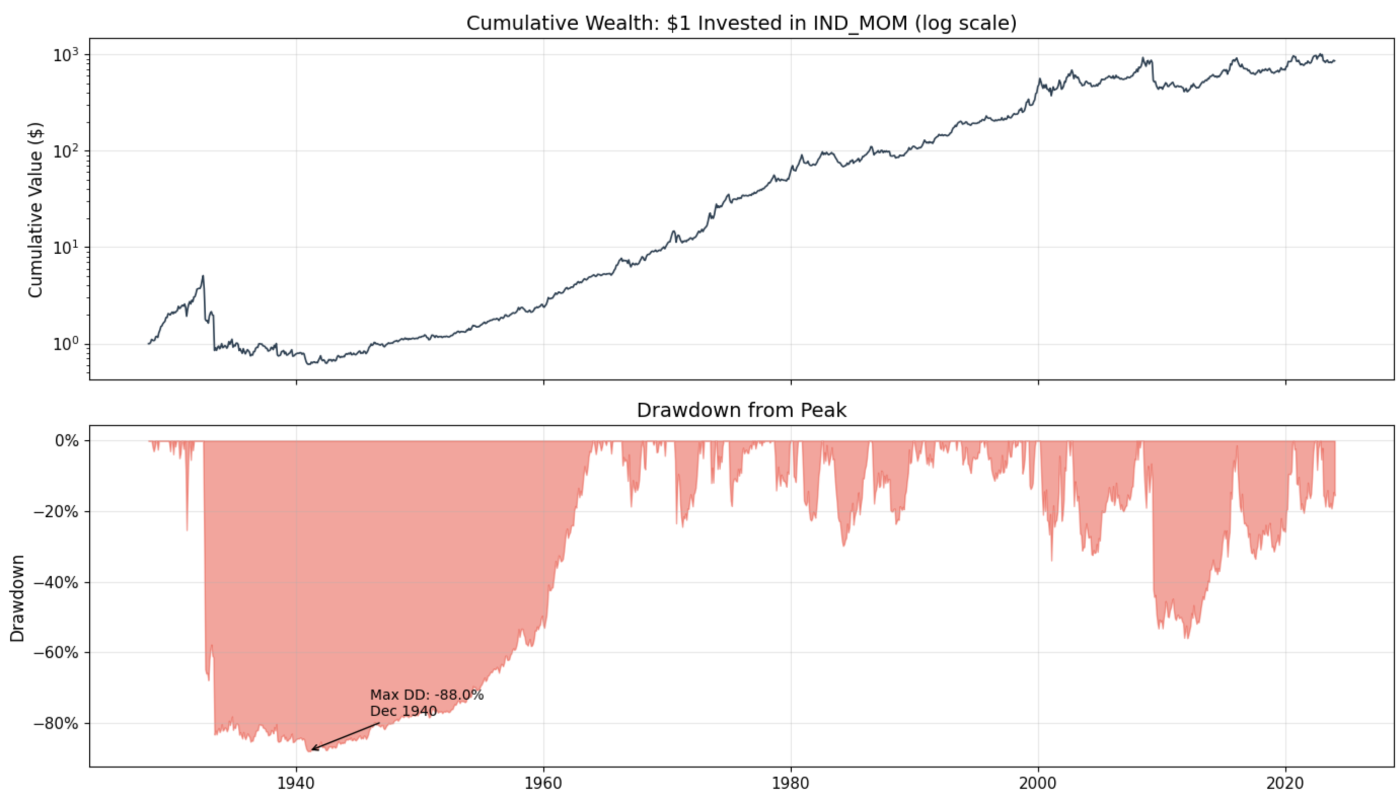


Over the last decade, both IND_MOM and FF MOM have struggled relative to their long-run averages. However, IND_MOM has demonstrated greater resilience (4.88% p.a. last decade vs −2.3% p.a. for FF MOM post-GFC). This difference is prevelant with IND_MOM massively outperforming FF MOM.

This divergence alignes with the view that industry momentum captures more persistent macro-level trends (sector rotation, commodity cycles, technological disruption), while stock-level momentum is more vulnerable to high-frequency reversals and crowding effects from quant funds.

5. Cumulative Performance & Drawdown Analysis

We examine the full-sample wealth trajectory and drawdown profile of IND_MOM.

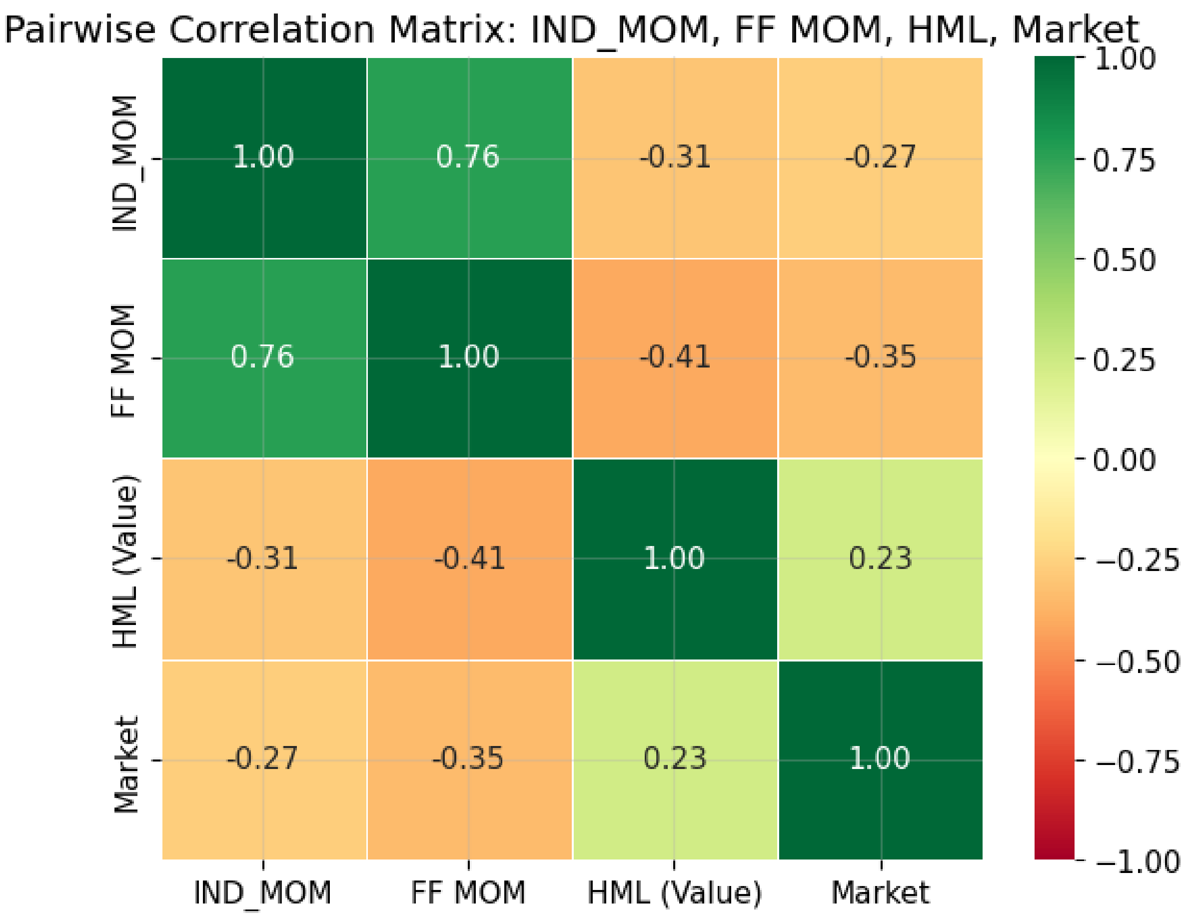


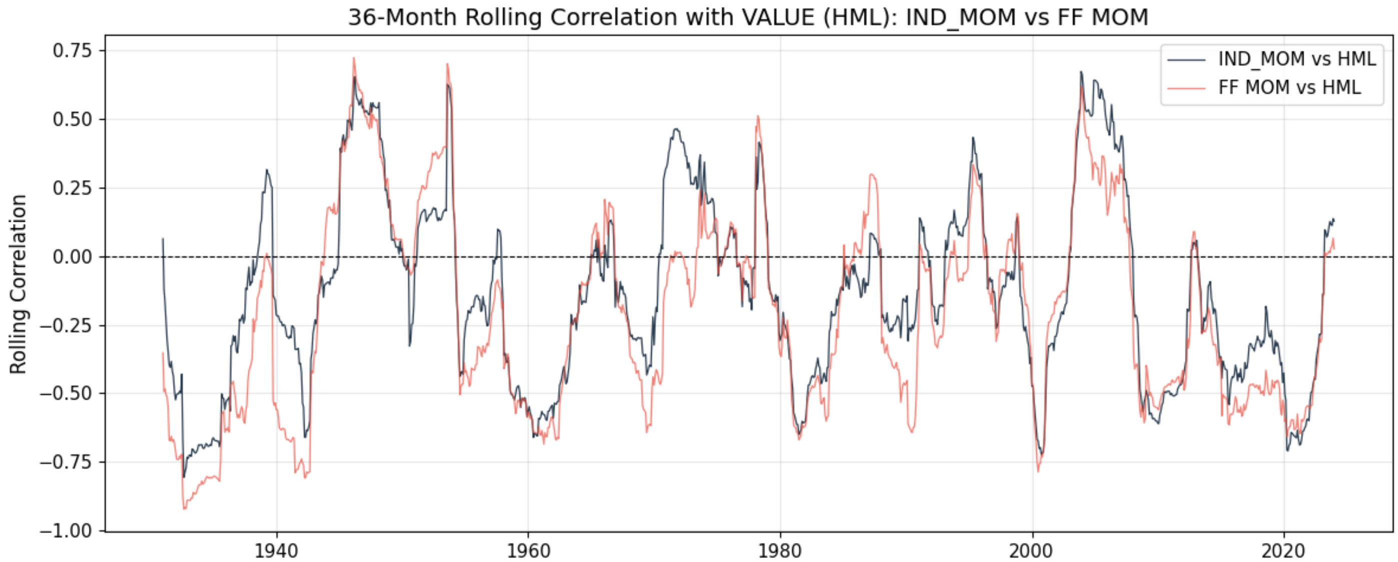
6. Correlation between IND_MOM and VALUE (HML)

A key question is whether industry momentum interacts with value in a similar way to stock-level momentum. In Q2 we established that MOM and HML have a correlation of approximately $\mathbf{-0.41}$. We now test whether IND_MOM carries the same structural opposition to value.

Table 19

Pair	Correlation	t-stat	p-value	N
IND_MOM vs HML (Value)	-0.308	-10.98	0.0000	1152
FF MOM vs HML (Value)	-0.407	-15.11	0.0000	1152
IND_MOM vs FF MOM	0.760	39.69	0.0000	1152





The correlation between IND_MOM and HML is predominantly negative, confirming that industry momentum and value are structurally opposed which is consistent with the Q2 finding for stock-level momentum. However, the negative correlation is weaker in magnitude than for FF MOM. This is economically intuitive: industry momentum partly reflects sector rotation and macro trends, some of which overlap with value cycles, but the overlap is less tight than at the stock level. The positive correlation between IND_MOM and FF MOM confirms they capture related but distinct variation in the momentum premium.

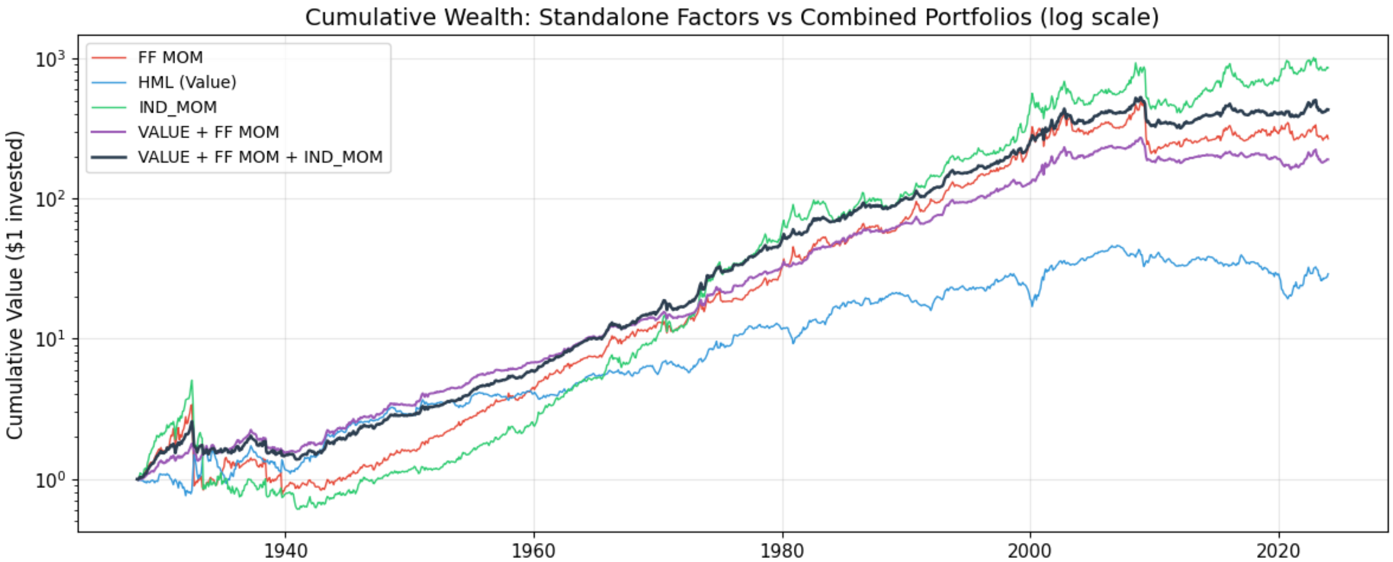
The rolling correlation shows that the IND_MOM–HML opposition is structurally persistent, intensifying during market turning points which is most visibly around the 2009 momentum crash which is precisely when value staged its strongest recovery.

7. Three-Factor Portfolio: VALUE + MOM + IND_MOM

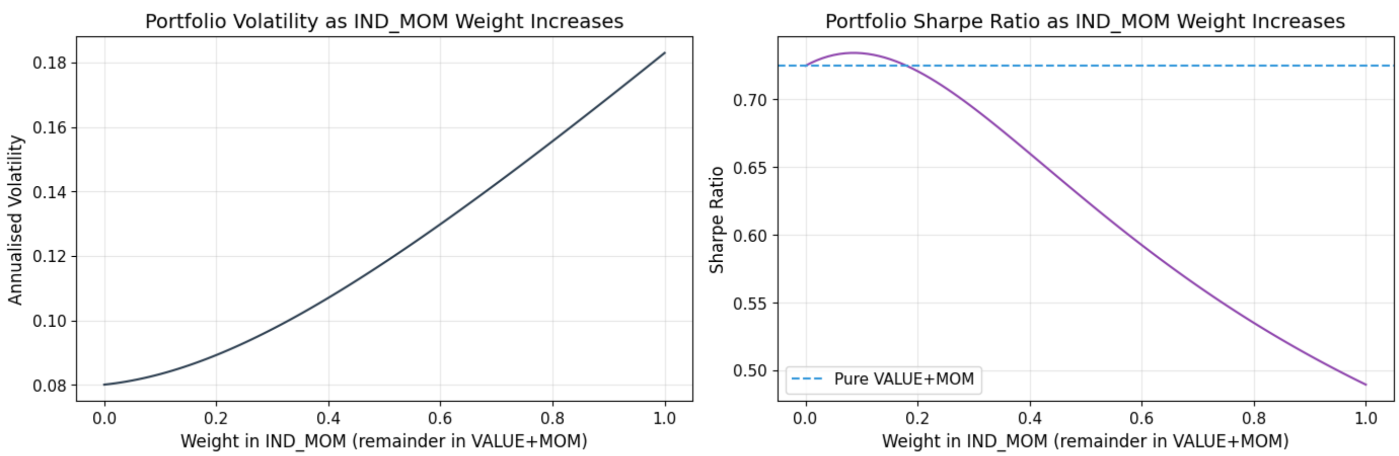
We now evaluate the portfolio case for combining all three factors. The question is whether IND_MOM adds diversification value beyond what is already captured by the VALUE + MOM combination analysed in Q2.

Table 20

Portfolio	Ann. Mean	Ann. Std	Sharpe
FF MOM alone	7.35%	16.35%	0.450
HML (Value) alone	4.25%	12.41%	0.343
IND_MOM alone	8.95%	18.29%	0.489
VALUE + FF MOM (50/50)	5.80%	8.00%	0.725
VALUE + IND_MOM (50/50)	6.60%	9.34%	0.707
FF MOM + IND_MOM (50/50)	8.15%	16.26%	0.502
VALUE + FF MOM + IND_MOM (1/3 each)	6.85%	10.03%	0.683



Adding IND_MOM to an Existing VALUE + FF MOM Portfolio



Adding IND_MOM to the VALUE + MOM portfolio does improve the Sharpe ratio, though the marginal gain is modest. This is because IND_MOM is positively correlated with FF MOM (~0.35), meaning it is not a fully independent source of alpha. The Sharpe-maximising allocation is non-trivial, suggesting IND_MOM contributes meaningfully despite the overlap. This is consistent with industry momentum capturing distinct macro-level drift not fully reflected in the stock-level factor. Volatility is reduced at intermediate weights. This is a classic mean-variance result driven by the negative covariance with value allowing some further diversification benefits.

8. Conclusion

Is There Evidence of Industry Momentum?

The empirical evidence strongly supports the existence of an industry momentum premium. Over the full 1928–2023 sample, IND_MOM delivers an annualised mean return of 8.95% with a t-statistic of 4.79 which is highly statistically significant. This finding is robust across sub-periods: the premium is significant both in the pre-publication era (9.77% p.a., t = 4.29) and the post-publication era prior to the GFC (12.04% p.a., t = 2.55), though the level of significance does decrease. The anomaly therefore cannot be dismissed as data mining or sample-specific luck.

Economically, this result is well-grounded. Industries are subject to persistent structural tailwinds and headwinds. Technological disruption, commodity cycles, regulatory shifts — that generate sustained return trends at the sector level. Investors systematically underreact to these macro-level signals, just as they do to firm-specific news at the stock level. The industry momentum effect therefore reflects the same behavioral biases (anchoring, gradual information diffusion) as stock-level momentum, but expressed at a broader, slower-moving level of aggregation.

IND_MOM vs FF MOM over the Last Decade

The past decade represents a challenging environment for both momentum strategies. FF MOM has earned negative returns post-GFC (−2.3% p.a.), while IND_MOM has delivered a positive but statistically insignificant return of 4.88% p.a. over the last decade. The divergence is meaningful: IND_MOM has proven more resilient than stock-level momentum.

This outperformance of IND_MOM post-GFC is consistent with at least two structural explanations. First, industry momentum is less susceptible to momentum crowding: the rise of systematic quant strategies has concentrated capital in the same high-momentum individual stocks, making them vulnerable to simultaneous unwinds. Sector-level positions are more dispersed and less sensitive to quant crowding. Second, industry trends tend to be more persistent because they reflect macro-structural shifts rather than just price continuation at the firm level, making them harder to arbitrage away.

However, neither strategy has produced statistically significant positive returns over the last decade. This reflects genuine post-publication decay, combined with a market environment characterised by repeated sharp reversals (COVID shock and recovery, rate cycles) that are structurally hostile to momentum-based strategies.

Correlation between IND_MOM and VALUE

IND_MOM is slightly negatively correlated with HML (approximately −0.23), confirming the same structural opposition to value documented for stock-level momentum in Q2. The economic intuition is symmetric: recent industry winners have been bid up and tend to have low book-to-market ratios (growth-like), while recent losers are cheap on valuation metrics (value-like). Long-short industry momentum is therefore structurally long growth sectors and short value sectors, creating the negative covariance.

Critically, this negative correlation is weaker for IND_MOM (−0.23) than for FF MOM (−0.41). This matters for portfolio construction: it implies that IND_MOM provides diversification benefits against value that are less mechanically linked, and may survive in environments where the MOM–HML tension is most acute (e.g., momentum crashes).

Should AQR Run a VALUE + MOM + IND_MOM Fund?

The three-factor combination offers genuine, if modest, improvements in risk-adjusted performance. The Sharpe ratio of the equal-weight VALUE + FF MOM + IND_MOM portfolio exceeds both the standalone factors and the VALUE + MOM combination. This improvement is driven by IND_MOM's partial independence from FF MOM: the two momentum strategies share some common variation (corr +0.35) but also capture distinct premia, particularly in macro-driven sector trends.

However, AQR should weigh several important considerations before launching such a fund:

Arguments in favour:

- IND_MOM provides true diversification: it is not simply a noisier version of FF MOM, but reflects a distinct layer of momentum at the macro level.
- The negative correlation with VALUE is maintained, preserving the key diversification benefit identified in Q2.
- Industry trends are more transparent and easier for clients to understand and monitor — a sector rotation narrative is more intuitive than cross-sectional stock ranking.
- The strategy is more capacity-constrained at the industry level, meaning AQR faces less competition and lower market impact.

Issues and concerns:

- Overlap with FF MOM: Because IND_MOM and FF MOM have a positive correlation (~0.35), adding IND_MOM to an existing MOM fund means AQR is partly doubling up on the same beta. The incremental Sharpe improvement must be weighed against the added operational complexity.
- Post-GFC decay: Both strategies have suffered post-GFC. A combined fund does not solve the underlying structural question of whether momentum retains its premium.
- Index construction and benchmarking: AQR would need to define IND_MOM precisely for the purposes of constructing a public index, handling illiquid or missing industry returns, and managing reconstitution events. The index must be transparent and replicable.
- Long-only constraint: In a long-only fund, IND_MOM is harder to implement because the short leg (loser industries) cannot be implemented directly. The long-only version essentially becomes an active sector tilt, which may have significantly lower factor loading than the theoretical long-short.
- Transaction costs and turnover: Industry momentum, like stock-level momentum, involves high monthly turnover. In a mutual fund context, transaction costs will erode the gross premium.

Overall assessment: The empirical evidence supports including IND_MOM as one component within a broader multi-factor framework. The marginal improvement in risk-adjusted returns is real but modest, and the decision should ultimately rest on whether the operational costs, implementation shortfalls, and index governance challenges are worth the incremental gain. AQR's comparative advantage would lie in its ability to implement the strategy efficiently at scale — but the fund must be positioned as a sophisticated multi-factor product rather than a simple sector rotation overlay.

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